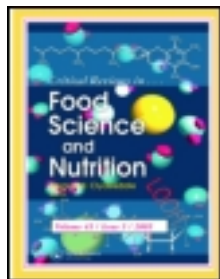


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Fish Flavor

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ABSTRACT: This article reviews features of flavor in three groups of fishes and summarizes them as follows: (1) fresh saltwater fish are nearly odorless because they contain a small quantity of volatiles; (2) freshwater fish give off pyrrolidine and earthy-odor compounds, which are responsible for their maturity and surrounding water pollution, and (3) euryhaline fish exhibit a variety of unsaturated carbonyls and alcohols derived from enzymatic and nonenzymatic oxidation of polyunsaturated fatty acids (PAs). These features are discussed, as are the effects of different enzymatic activities on PA oxidation and the effects of pH on mechanisms of formation of the volatiles. The monotonous volatile constitution of saltwater fish is likely caused by an unknown antioxidation system restraining the fish from oxidizing. The variety of constitution of euryhaline fish, especially that of anadromous fish under spawning conditions, could result from the loss of that system. The thermal environments of heated foods are also reviewed. The basic environment of fish, which allows the formation of the flavor compounds, is discussed to confirm the volatiles found in unheated fish.

KEY WORDS: saltwater fish, volatile compounds, pH.

I. INTRODUCTION

With the development and the diffusion of a gas chromatograph equipped with a capillary column and a mass spectrometer, the amount of seafood volatile components identified has increased. Identifiable compounds have been compiled by Maarse and Visscher,¹ and their influence on seafood flavors is discussed.

Until now, many studies on seafoods have focused on the chemical changes during storage to prevent deterioration and spoilage. Much information has accumulated concerning seafood volatiles, especially low-molecular-weight volatile compounds involving the so-called off-flavor components. Ikeda² summarized in his review those principal compounds from a variety of marine products, and later Yoshinaka³ and Tokunaga⁴ added more detail.

Before the 1980s, most of the studies on seafood volatiles concentrated on ways to suppress

or remove an unpleasant fishy odor so that seafood could be used more widely for its nourishing protein quality. The most common reason for consumer rejection is the unacceptable flavor, even though fish contain healthy oils and high-quality proteins. The sensory aspects of seafood are extremely visible to the consumer and are essential for consumer satisfaction.⁵ A fishy odor seems to have prejudiced some consumers against eating seafood. Because of their faint odor, kamabokos and fish sausages, manufactured through surimi from fish muscle, have been much favored by such people. These fish products have been carried to far inland areas where foods rich in protein are seldom abundant.

Modern commercial cold treatment systems and home refrigerators have helped to decrease seafood's fishy odor. Off-flavors in fish may be caused by its rapid assimilation with taints in water and by auto-oxidative processes during storage, as described by Reineccius.⁵ As the fishy

odor seems to be the more troublesome problem, research on the overall volatile components of seafoods has increased. Many researchers have developed enzymatic reactions of lipoxygenases in fish tissue with polyunsaturated fatty acids (PAs) for characterizing many carbonyl and alcohol compounds^{6,7} that contribute the distinctive plant-like odors to some freshwater fishes^{8,9} and oysters.¹⁰ Furthermore, the enzymatic reactions reveal some mechanisms of formation for those aroma volatiles derived enzymatically and nonenzymatically from PAs.

Many good reviews on seafood flavors, along with the above ones, have been published as follows: for fresh fishes,^{11,12} fishes and other seafoods,^{13,14} dried bonito (Katsuobushi),¹⁵ prawns and lobsters,¹⁶ shellfish,¹⁷ processed seafoods,¹⁸ shrimps,¹⁹ crabs and crayfish,²⁰ seaweeds,^{21–23} and off-flavors.^{5,24,25} Because those reviews were devoted to volatile components in each area, this article will focus on the principal volatiles of fishes together with the effect of pH on their formation and odor quality.

The pH of food strongly affects its flavor, as seen here and in the classic review by Schultz et al.²⁶ The significant effect of pH on the odor of fish was pointed out by Jones.²⁷ pH has affected the mechanisms of biosynthetic formation for fish volatiles¹² and odor of trimethylamine (TMA).²⁸ Further, pH also affects the generation of some small molecular weight compounds: for example, hydrogen sulfide (H₂S) from heated meats²⁹ and proteins,³⁰ dimethyl sulfide (DMS) from marine products,^{31,32} and TMA from fish muscle.³³ Generally, simple compounds that have low boiling points (bp) produce more complex flavor compounds through their interactions, such as *S,N*-heterocyclic compounds (sulfur and nitrogen atoms in the same ring). Aroma arising from foods, of which distillates are ordinarily submitted to gas chromatography-mass spectrometry (GC-MS) analysis for research on the volatiles, must be influenced by pH as well. On Maillard and other model reactions, a lot of pH data accumulated from the reaction conditions have backed the postulation on formation mechanisms for the resultant compounds.^{34–39} These studies demonstrated that pH plays an important role in producing many heterocycles and other flavor compounds.⁴⁰ Nevertheless, pH values of distillates

from natural products have been obscure, except for some limited seafood distillates.^{41–44} This review, therefore, also discusses the relationship between pH and fish flavor.

II. FEATURES OF FISH FLAVOR

A. Volatile Components of Saltwater Fish

Fresh saltwater fish is nearly odorless.⁴⁵ Fresh saltwater fish, just after being caught in the ocean, give off a faint odor. This is a feature that belongs to all live saltwater fishes such as Pacific mackerel, bonito, sardine, and Alaska pollack.⁴ From the indistinct odor of fresh saltwater fish, several volatile components have often been detected:^{3,4} nitrogen-containing compounds (N-compounds) such as ammonia, TMA, and dimethylamine (DMA); acetic, formic, and propanoic acids; and the sulfur-containing compounds (S-compounds) DMS, H₂S, and methanethiol.

Although each compound has been found in a variety of foods, the foods reported as having all the nine volatiles were unexpectedly few. Saltwater fish, cheese, pork, and beef were compiled from 200 items of food.⁴⁶ Because it must be rather difficult to identify some light volatiles by GC-MS analysis only, the number of food items containing such compounds may be underestimated because limited detection of TMA in foods restricts the number. The small number allows TMA the most characteristic volatile component among the nine compounds for fresh saltwater fishes. However, the combination of TMA and other compounds contributes more to saltwater fish odors than TMA alone: both dilutions of TMA and DMS possess dried (or boiled) fish-like and seashore product-like odors, respectively, but their mixture has a different odor associated with seafood flavors. Further, other combinations of TMA and acids change the odor quality, as shown in Table 1.²⁸ The change in odors indicates that the relationship between an odor and pH is important for odor evaluation. The combination of TMA and the acids, as well as that of TMA and DMS, has been applied to the manufacturing of seafood flavors.

Several volatile compounds detected in some saltwater fishes are summarized in Table 2,¹¹ to-

TABLE 1
Changes of Odor in the Admixtures of TMA and Volatile Fatty Acids (FA)

Test solution TMA (mg/ml) + FA (mg/ml)		pH	Amount of evaporated TMA (mg/ml)	Sensory Test
Acetic acid				
0.031	0	9.4	0.023	TMA odor
0.031	0.020	7.3	0.002	Fishy, boiled fishy
0.031	0.050	6.5	0.002	Fishy, unpleasant
0.031	0.100	4.7	0	Acetic acid
Propanoic acid				
0.035	0	9.4	0.026	TMA
0.035	0.008	7.8	0.014	Fishy, canned fishy
0.035	0.040	6.8	0.007	Fishy, unpleasant
0.035	0.200	4.8	0	Acid
Butanoic acid				
0.035	0	9.4	0.026	TMA
0.035	0.005	9.0	0.006	Fishy, rancid
0.035	0.010	7.5	0.005	Fishy, unpleasant
0.035	0.050	7.1	0.004	Unpleasant

From Kikuchi et al.²⁸ With permission.

gether with their odor thresholds in water. Table 2 supports the fact that neither characteristic carbonyl nor alcohol compounds were detected in fresh saltwater fish.⁴ Some treatments (storage and processing) given in Table 2 enhanced generation of various volatile components, suggesting that enzymatic or nonenzymatic oxidative reactions occurred during the treatments.

B. Volatile Components of Freshwater Fish

In contrast to saltwater fish, many freshwater fish residing in ponds, rivers, and lakes, such as carp, catfish, and even cultured trouts, have an earthy odor.⁴ The odors of fresh carp and salmon have been estimated for a long time to be derived from piperidine⁴⁷ and its complex of ethanal (possibly, 1,1-bispiperidino-ethane).⁴⁸ The study concluded that the combination of piperidine and ethanal was responsible for the earthy odor and that combination plus TMA was responsible for the

faint odor of saltwater fish.⁴⁸ Recently, piperidine and pyrrolidine were determined in fresh carp muscle and in 12 various products of canned salt- and freshwater fish:⁴⁹ piperidine was detected in only two canned red salmon (0.05 mg/100 g) and pond smelt (0.02 mg/100 g) in small amounts, but not in the carp muscle, whereas pyrrolidine was found in larger amounts in all tested fishes. Thus, piperidine is not a common component among ordinary fresh- and saltwater fishes.

However, because the reaction products of piperidine with other compounds have remained obscure, an attempt to develop the model reactions by using a capillary GC-MS is much expected for characterizing more fish volatiles. This is necessary because many *N*-substituted piperidine derivatives have fishy odors, and also because piperidine has an active property that reacts with aldehydes and alcohols.

High piperidine contents (6.1 mg/100 g muscle and 4.6 mg/100 g skin) were reported in a salmon just after spawning, piperidine levels increasing rapidly as spawning approached.²⁵ Due to its off-

TABLE 2
Identified Compounds from Saltwater Fishes and Their Odor
Thresholds in Water (ppb)

Compound	Fresh fish ^a	Cold stored fish ^b	Dried fish ^c	Cooked fish ^d	Odor threshold in water (ppb)
Carbonyls					
Formaldehyde		+	+	+	
Ethanal		+	+	+	15 ^e 109 ^e 120 ^e
Propanal		+	+	+	9.5 ^f
2-Methylpropanal		+		+	0.1 ^e 2.3 ^e 10 ^e
Other aldehyde		+	+	+	
Acetone		+		+	
Butanone		+		+	
Other ketone		+			
Bases					
Ammonia	+		+	+	110,000 ^g
Dimethylamine	+	+	+	+	30,000 ^g
Trimethylamine	+	+	+	+	600 ^g
Other amine	+			+	
Piperidine				+	
Acids					
Formic acid	+			+	1,000,000 ^g
Acetic acid	+	+	+	+	34,200 ^g
Propanoic acid	+		+	+	32,800 ^g
Butanoic acid	+		+	+	3,000 ^g
Other acid	+		+		
Sulfur compounds					
Hydrogen sulfide	+	+	+	+	10 ^e
Methanethiol	+	+		+	0.02 ^e
Dimethyl sulfide	+	+		+	0.33 ^e 1 ^e 10 ^e
Dimethyl disulfide		+			0.16 ^e 1.2 ^e 12 ^e
Miscellaneous					
Alcohols		+		+	
Hydrocarbons	+	+		+	

^a Bonito, Pacific mackerel, a sea bream (*Pristipomoides sieboldi*), and Pacific herring.

^b Pacific halibut and Pacific cod.

^c Pacific cod and Alaska pollack.

^d Bonito, Pacific mackerel, jackmackerel, sardine, the bream, and Pacific herring.

^e Reference 125.

^f Reference 3.

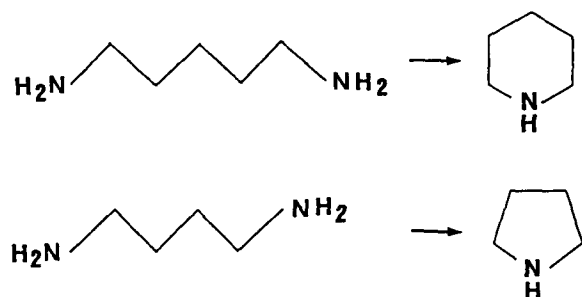
^g Reference 28.

From Hirano¹¹, except for the threshold values e-g. With permission.

flavor, salmon has been rejected for a man-food, and of no use except for animal-feed or mature after taking out the roe as a delicacy. Piperidine has frequently been found in great quantity (3 to 4 mg/100 g muscle, 13 mg/100 g skin) in large grass carp.²⁵ These findings suggest that piperidine concerns the spawning or the succeeding mortality for the salmon and the grass carp.

Piperidine and pyrrolidine are biogenetical products, by cyclization, of 1,5-diaminopentane (cadaverine) and 1,4-diaminobutane (putrescine), respectively.^{47,49} The narrow and wide distribution of piperidine and pyrrolidine is almost consistent with that of two diaminoalkanes, respectively: 1,5-diaminopentane was found in loach alone (2 mg/100 g) among carp, rainbow trout, eel, lo-

ach, and snake-head, whereas 1,4-diaminobutane was found in carp, rainbow trout, and snake-head (about 1 mg/100 g).⁵⁰ These small amounts of the diaminoalkanes cannot contribute to freshwater fish odors, because of their weak odors.²⁷



SCHEME 1. Formation of piperidine and pyrrolidine.

The earthy odor of freshwater fishes is responsible for the metabolites of inhabitant microorganisms, geosmin and 2-methyl-*iso*-borneol;^{3,51,52} mold-like bacteria (actinomycetes) and blue-green algae produce larger quantities of these compounds as pollution of freshwater areas increases. In particular, geosmin possesses a powerful earthy odor with a very low threshold at 0.01 to 0.02 ppb in water.⁵³ Many freshwater fishes residing in ponds and lakes may have these compounds, especially geosmin, in very small amounts as their characteristic flavor components. The presence of these compounds may not be related to pollution, because freshwater fishes are well known for their earthy odor, even if there is no water pollution.

C. Volatile Components of Euryhaline Fish

Only the euryhaline fish in a freshwater area, such as salmon, trout, and smelt, which are able to live principally in both water areas, were reported to smell like cucumbers and melons.^{8,13}

As Table 3 shows,⁵⁴ Pacific Ocean coho salmon still residing in the sea near the mouth of a river at the prime season (A in Table 3) had smaller amounts of volatiles and a simpler volatile constitution than when it entered fresh water for spawning (B in Table 3). In salmon A, ethanol and five C₈-compounds were found. The differences in the levels of volatile aroma compounds between salmon A and salmon B indicated that

some physiological change occurred during osmoregulation and/or preparation to spawn.⁵⁴ However, the gap in levels of volatile compounds, as given in the partial volatile constitution in Table 3, between Great Lake coho salmon growing to prime (C in Table 3) and maturing well to spawn in fresh water (D in Table 3), was smaller than the gap of osmoregulation between salmon A and B. The levels recorded for salmon C and salmon D were similar for several other freshwater salmon.⁵⁴ Therefore, the dramatic gap between A and B must have been derived from a combination of osmoregulation and spawning conditions.

Salmon B consumes much of its energy for changing its osmoregulation and preparing for spawning. The large production of carbonyl and alcohol compounds by salmon B might, therefore, be caused by its inability to prevent oxidation of PAs, consequently allowing many volatiles to be formed and released. This speculation is based on the fact that there is little detection of carbonyl and alcohol compounds in saltwater fishes and other ordinary freshwater fishes. The physiological significance of the formation of these compounds for salmon B is unknown.

The volatile constitution of salmon A is similar to that of many saltwater fishes. Salmon living in the sea and coming to the coasts have been favored for Japanese diets because of their suitable flavor, possibly because of the small quantities of the five unsaturated C₈-compounds listed in Table 3 (1,5-octadien-3-ol, 1-octen-3-ol, 2,5-octadien-1-ol, 1,5-octadien-3-one, and 1-octen-3-one).

D. Carbonyl and Alcohol Compounds in Fish

1. Euryhaline Fish

The difference in volatile constitution between salt- and freshwater fishes was proven by a comparison of several constituents of fish extracts: 4 species of saltwater fish (cod, petrole sole, ocean perch, and haddock) and 12 species of freshwater fish. As shown in Table 4,⁵⁵ only four compounds (hexanal, 1-octen-3-ol, 1,5-octadien-3-ol, and 2,5-octadien-1-ol) were obviously detected as mutually common volatiles of both-water fishes among 14 identifiable compounds (eight carbonyls and six alcohols). The monotonous tone was almost

TABLE 3

Volatile Aroma Compounds Identified in Pacific Coho Salmon Residing in the Sea, Great Lake Coho Salmon in Fresh Water, and Those in Their Spawning Conditions in Fresh Water

Compound	Concentration in coho salmon extract (ppb)					
	Pacific Ocean coho				Great Lake coho	
	A Prime in salt water	B Spawning in fresh water	A'	B'	C Prime in fresh water	D Spawning in fresh water
Ethanol	19.5	24.8				
2-Butanol	— ^a	6.5				
Hexanal	—	8.3	—	8.3	0.52	Trace
1,3,5-Octatriene	—	0.3				
1,3,5-Octatriene	—	0.3				
2-Heptanone	—	0.5				
2-Pentenal	—	1.0				
1-Penten-3-ol	—	0.5				
Heptanal	—	0.8				
2-Hexenal	—	0.5				
1-Pentanol	—	0.3				
4Z-Heptenal	—	0.3				
3-Octanone	—	0.3				
Octanal	—	0.3				
1-Octen-3-one	Trace	0.5	Trace	0.5	Trace	Trace
Cyclopentanol	—	0.5				
Ethyl heptanoate (an internal standard)	45	45				
1-Hexanol	—	0.3				
1,5-Octadien-3-one	Trace	Trace	Trace	Trace	Trace	0.03
3-Hexen-1-ol	—	0.3				
3-Octanol	—	1.0				
2-Nonanone	—	0.3				
Nonal	—	0.3				
1-Octen-3-ol	1.0	18.8	0.4	18.8	8.0	5.8
1-Heptanol	—	0.3				
Acetic acid	—	3.3				
1,5-Octadien-3-ol	3.3	37.1	1.1	37.0	2.5	1.6
2E-nonenal	—	Trace	—		—	0.16
2E,6Z-nonadienal	—	0.3	—		—	0.08
2-Undecanone	—	0.3				
2-Octen-1-ol	—	3.5	—	3.5	3.3	1.5
1-Nonanol	—	0.3				
2,5-Octadien-1-ol	0.1	7.8	0.1	7.8	1.0	0.5
6-Nonen-1-ol	—	Trace	—	Trace	—	Trace
3,6-Nonadien-1-ol	—	0.3	—	0.3	—	Trace
2,4-Decadienal	—	0.3				
Unknown	—	0.3	—	0.3	—	0.5

Note: (A) Prime, saltwater Pacific coho salmon and (B) its spawning-condition in fresh water; (C) prime, freshwater Great Lake coho salmon and (D) its spawning condition in fresh water.

^a Not detected.

From Josephson et al.⁵⁴ With permission by combination of three tables on salmon coho volatiles.

TABLE 4
Occurrence of Volatiles in a Variety of Fresh Species

Species	C ₆ -anal	2E-C ₆ -enal	2-C ₆ -enal	2(Z?)-C ₆ -en-1-ol	1-C ₈ -en-3-one	1-C ₈ -en-3-ol	1,5(Z?)-C ₈ -dien-3-one	1,5(Z?)-C ₈ -dien-3-ol	2,5(Z,Z?)-C ₈ -dien-1-ol	2,3-C ₈ -dione	2E-C ₈ -enal	2E,6Z-C ₈ -dienal	6(Z?)-C ₈ -en-1-ol	3,6(Z,Z?)-C ₈ -dien-1-ol
Freshwater fish														
Lake white fish ^a	+	+	+	+	+	+	+	+	+	+	+	+	+	+
Ciscoe ^b	+	+	+	+	+	+	+	+	+	+	+	+	+	+
Muskellunge ^c	+	+	+	+	+	+	+	+	+	+	+	+		+
Northern pike ^d	+	+	+	+	+	+	+	+	+	+	+	+		+
Smelt ^e	+	+	+	+	+	+	+	+	+	+	+	+	+	+
Emerald shiner ^f	+	+		+	+	+		+	+	+	+	+		
Walleye pike ^g	+				+	+	+	+	+					
Sauger ^h	+				+	+	+	+	+					
Yellow perch ⁱ	+				+	+	+	+	+					
Rainbow trout ^j	+				+	+	+	+	+					
Bluegill ^k	+		+		+	+	+	+	+					
Black crappie ^l	+		+		+	+	+	+	+					
Saltwater fish														
Cod ^m	+					+		+	+					
Petrole sole ⁿ	+					+	±	+	+					
Ocean perch ^o	+					+		+						
Haddock ^p	+					+	±	+						

Note: +, denotes occurrence; ±, trace; determined by odor assessment from packed column equipped with an effluent splitter.

^a *Coregonus clupeaformis*.

^b *Coregonus artedii*.

^c *Esox masquinongy*.

^d *Esox lucius*.

^e *Osmerus mordax*.

^f *Notropis atherinoides*.

^g *Stizostedion vitreum*.

^h *Stizostedion canadense*.

ⁱ *Perca flavescens*.

^j *Salmo gairdneri*.

^k *Lepomis macrochirus*.

^l *Pomoxis nigromaculatus*.

^m *Gadus morhua*.

ⁿ *Eopsetta jordani*.

^o *Sebastes marinus*.

^p *Melanogrammus aeglefinus*.

From Josephson et al.⁵⁵ With permission.

the same as that of the sea salmon in Table 3. The four common compounds, which have rather earthy and sober odors (Table 5),¹³ have relatively higher thresholds (T; 4.5, 10, 10, and 10 ppb, respectively) than other compounds commonly found in freshwater fish, such as 1-octen-3-one (T, 0.09 ppb) and 1,5-octadien-3-one (T, 0.001 ppb). Thus, saltwater fish have a more monotonous volatile constitution and more moderate odors than freshwater fishes. Generally, unsaturated carbonyl and alcohol compounds are recognized as having powerful plant-like green odors (Table 5), so that the freshwater fishes give off the distinct odors.

All the listed freshwater fish in Table 4, except the obscure emerald shiner, are considered

to be euryhaline fish: the anadromous (lake white fish, ciscoe, smelt, and rainbow trout), the catadromous (sauger), and other fish that can be found in salt and brackish water, in addition to fish living in fresh water (muskel-lunge, northern pike, walleye pike, yellow perch, bluegill, and black crappie). The land-locked anadromous fish have the characteristic odors, together with ordinary salmon and trout. The odors are quite different from those of common freshwater fish as described above. Therefore, this group of fish is considered the euryhaline fish in freshwater.

Almost all the compounds in Tables 3 to 4 were found not only in the euryhaline fish but also in some saltwater fish oils, such as tuna oil,⁵⁶ cod

TABLE 5
Enzymatically Derived Volatile Aroma Compounds from Fatty Acids
Identified in Freshly Harvested Seafood

Compound	Aroma quality	Odor threshold value (ppb)
2-Pentenal	Green, apple-like	n.a. ^a
1-Penten-3-one	Solvent-like	1.3
1-Pentanol	Wine-like, ethereal	120
1-Penten-3-ol	Grassy, green	400
Hexanal	Green, aldehyde	4.5
2E-Hexenal	Green, stinkbug	17
3Z-Hexenal	Green, apple-like	0.25
Hexanol	Green, fatty	500
3Z-Hexen-1-ol	Green, leaf-like	70
3-Octanone	Resinous, ethereal	50
1-Octen-3-one	Mushroom	0.09
1,5Z-Octadien-3-one	Geranium leaves	0.001
3-Octanol	Mushroom	18
1-Octen-3-ol	Mushroom	10
1,5Z-octadien-3-ol	Earthy, mushroom	10
2Z-octen-1-ol	Fatty, rancid	40
2E,5Z-octadien-1-ol	Earthy, mushroom	10
2E-nonenal	Cucumber, cardboard	0.08
3Z-nonenal	Watermelon	0.08
6Z-nonenal	Green melon rind	n.a.
2E,6Z-nonadienal	Cucumber peel	0.01
3Z,6Z-nonadienal	Watermelon	n.a.
2Z-nonen-1-ol	Green, waxy melon	n.a.
6Z-nonen-1-ol	Green, waxy melon	n.a.
2E,6Z-nonadien-1-ol	Dry green, cucumber	n.a.
3Z,6Z-nonadien-1-ol	Cucumber, melon rind	10
1,5Z-undecadien-3-ol	n.a.	n.a.
1,5Z,8Z-undecatrien-1-ol	n.a.	n.a.

^a n.a. = not available.

From Josephson.¹³ With permission.

liver oil,⁵⁷ and menhaden oil.⁵⁸ The widely and mutually common variety of volatile constitution among them suggests that oil processing thermally destroys an antioxidative system in saltwater fish tissue, and consequently releases common oxidative products of PAs similar to those of the euryhaline fish.

2. Formation of Carbonyl and Alcohol Compounds

Three principal mechanisms for the formation of fresh fish aroma volatiles were summarized.^{8,13} Carbonyl and alcohol compounds were the result of oxidative reactions with ω -3 PAs,

eicosa-5,8,11,14,17-pentaenoic acid (EPA), docosa-4,7,10,13,16,19-hexaenoic acid (DHA) in the fish lipids, as shown in the following three figures. Figure 1,⁸ biosynthetic reactions of EPA with lipoxygenases and hydroperoxide lyases produce unsaturated carbonyls and alcohols, such as 3Z,6Z-nonadienal, 1,5Z-octadien-3-one, and 1-penten-3-ol; Figure 2,¹³ autooxidation of EPA produces other types of unsaturated carbonyls, such as 2E,4Z-heptadienal and 3E,5Z-octadien-2-one; finally, Figure 3,¹³ spontaneous retro-aldol condensation converts higher unsaturated aldehydes like 2E,6Z-nonadienal to the lower fatty aldehydes, 4Z-heptenal and C₂-C₃ saturated aldehydes. Also, similar biosynthetic pathways of C₆ and C₉-compounds have been proposed from li-

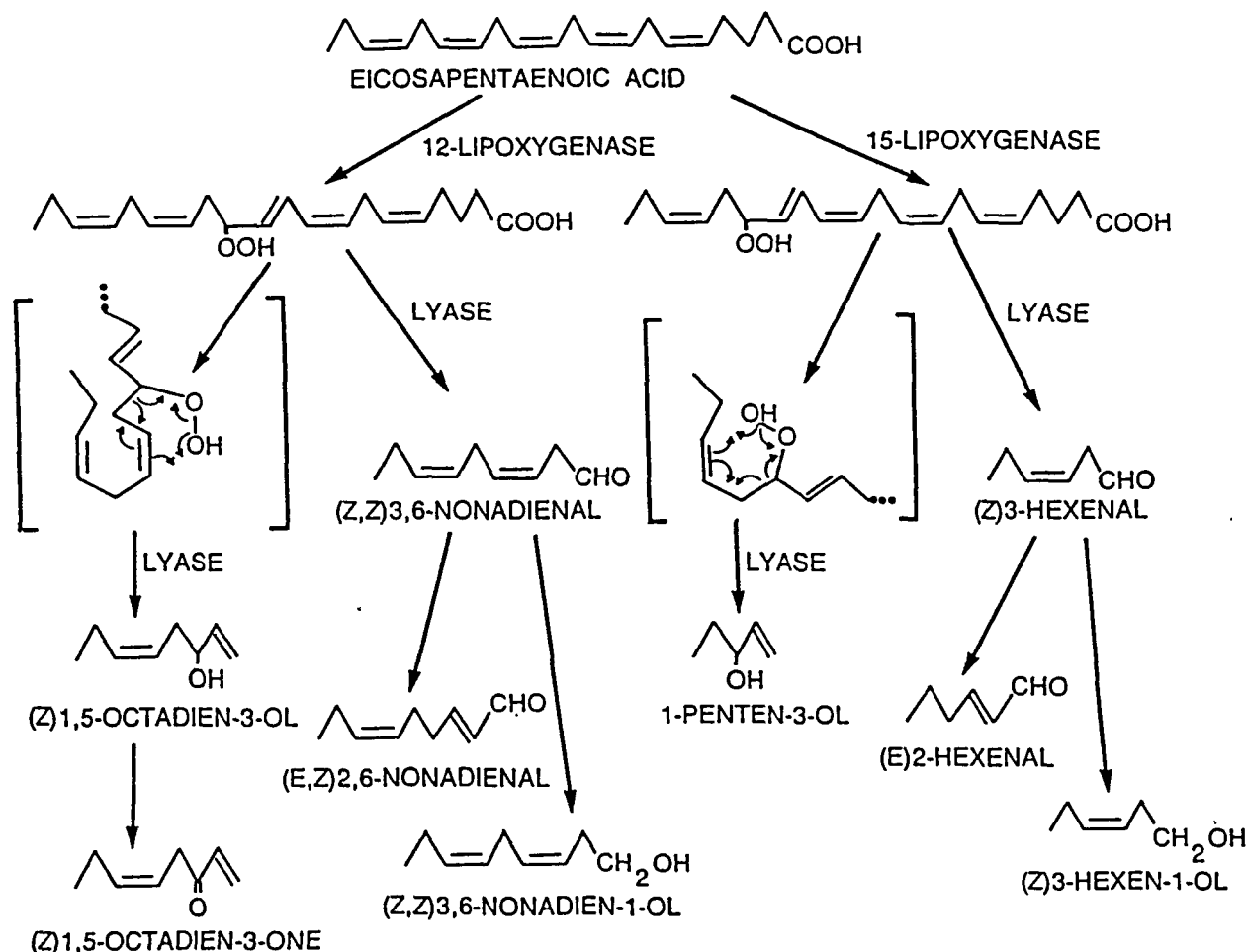


FIGURE 1. Proposed mechanism for the enzymatic biosynthesis of some fresh volatile seafood flavor compounds from eicosapentaenoic acid. (From Josephson and Lindsay.⁸ With permission.)

noleic acid (LA; 9Z,12Z-octadecadienoic acid) and linolenic acid (9Z,12Z,15Z-octadecatrienoic acid) for the generation of terrestrial plant⁵⁹ and seaweed volatiles.²³

Due to the cucumber- and watermelon-like odors, ayu smelt (*Plecoglossus altivelis*) living in

Thus, the euryhaline fish in a freshwater area give off various characteristic aroma volatiles

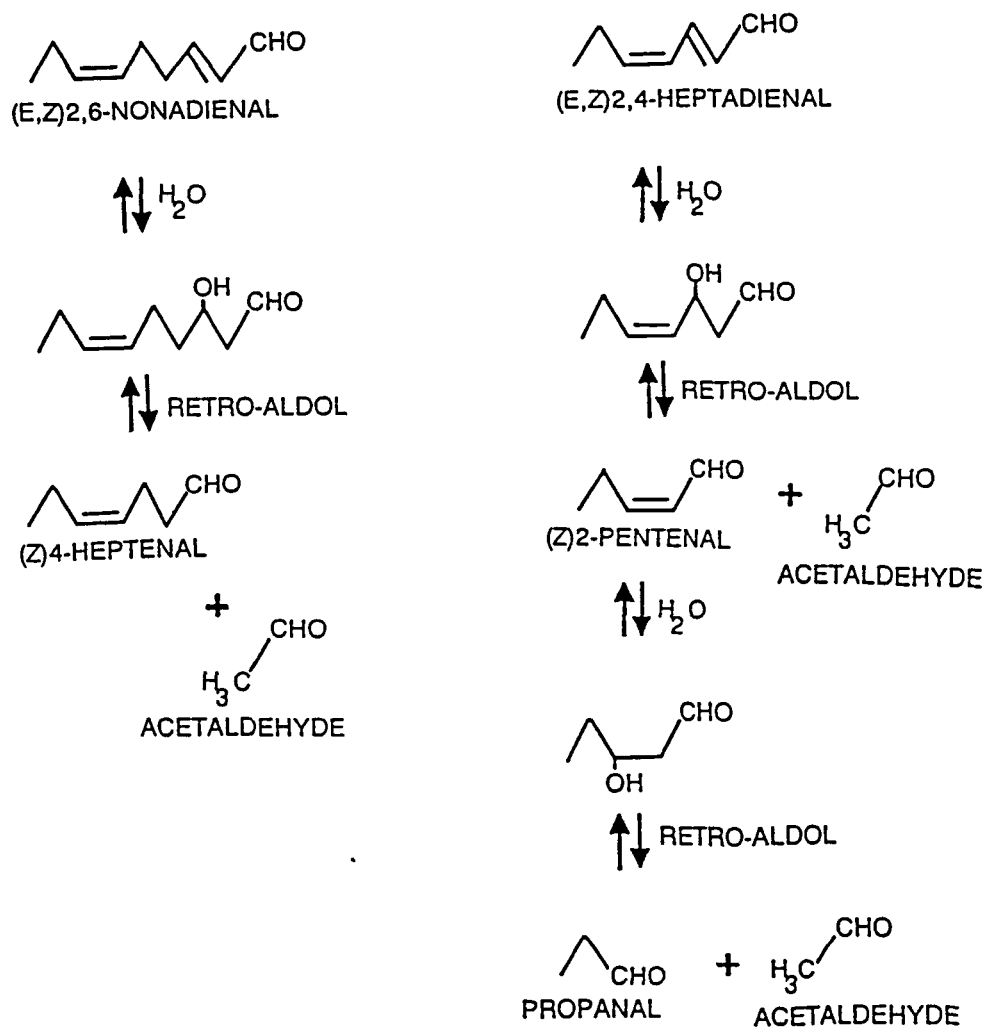


FIGURE 3. Mechanism for the formation of 4Z-heptenal from 2E,6Z-nonadienal and 2E-pentenal and of propanal from 2E,4Z-heptadienal via α/β double-bond hydration and retro-aldol condensation. (From Josephson.¹³ With permission.)

(Tables 3 to 5). Fresh saltwater fish only gave off a few simple volatiles.^{45,67} The volatiles arising from fresh muscle with skin of sardine heated in a temperature range of 22 to 99°C were also simple, 1-penten-3-ol being the main volatile (Figure 5⁶⁸). Even from the headspace of roasted sardine, 1-penten-3-ol was found along with five low fatty aldehydes and ethanol.⁶⁹ The alcohol has a pungent mustard aroma with a tingling aftertaste,⁷⁰ and a grassy-green note (Table 5). These odor descriptions almost agree with the observation that saltwater fish have a green-metallic odor.²⁷ Because the alcohol is a common volatile to fresh-

water fishes and has a relatively high odor threshold value (400 ppb in Table 5), it is unlikely that the alcohol contributes to the freshwater fish aroma, as described by Josephson.¹³ However, it must be a characteristic aroma compound for sardine as Figure 5 shows. Thus, the odor of saltwater fish is monotonous and faint, suggesting that low-molecular-weight compound such as TMA and DMS seem more significant for saltwater fish odors than the compounds listed in Table 5.

Almost no odor in fresh saltwater fish can be deduced from diets of fresh fish muscle "sasimi"

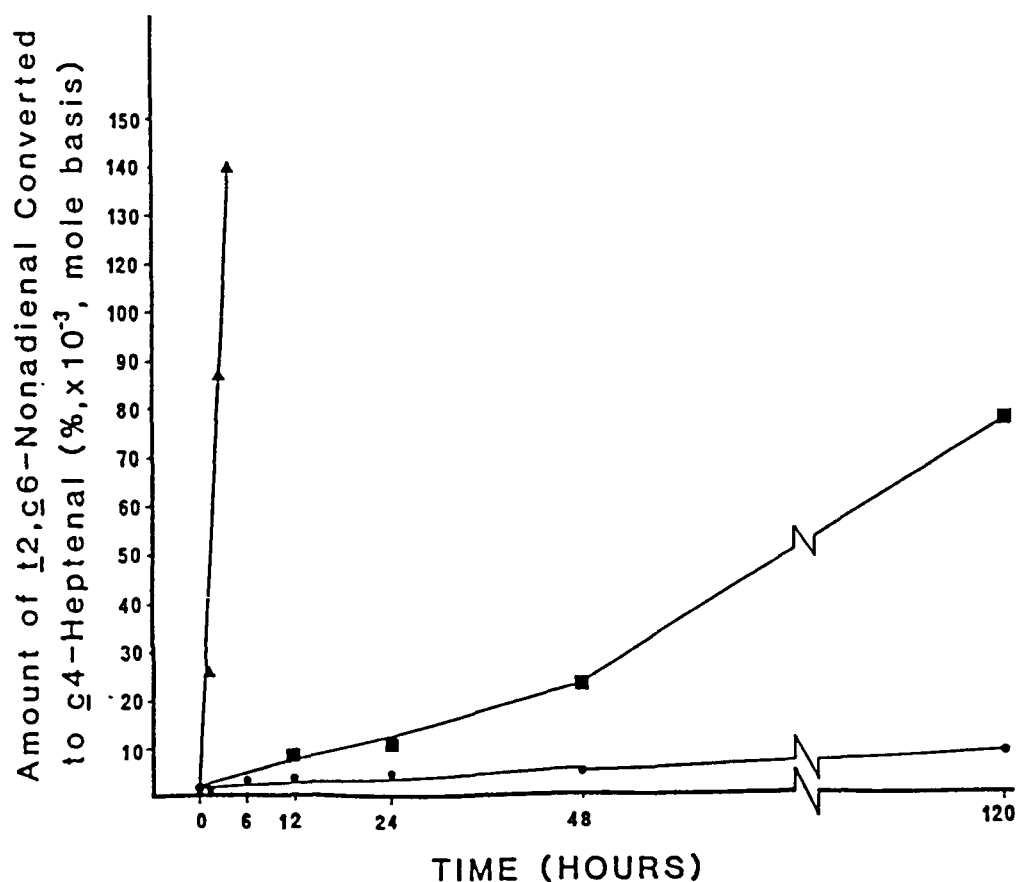


FIGURE 4. Effect of pH on the formation of 4Z-heptenal from 2E,6Z-nonadienal in aqueous model systems at 21°C (▲, pH 11.1; ■, pH 7.4; ●, pH 2.8). (From Josephson and Lindsay.⁶³ With permission.)

and “susi” that are eaten in the raw after removal of the skin and internal organs. Because of the odorlessness, sniffing at such fish cannot correctly distinguish one type of fish from the other. This is not true for salmon and trout. Also, fresh skins must be hardly identifiable because they emit commonly weak and green-metallic odors.

Recently, the lipoxygenase of saltwater fish was reported.^{61,62} The lipoxygenase isolated from sardine skin remarkably oxidized LA and its methyl ester more efficiently than it did EPA, DHA, and ARA, as exhibited in Table 6.⁶¹ Its characteristic 13(S)-lipoxygenase activity, due to the production of 13(S)-hydroperoxy-9Z,11E-octadecadienoic acid from LA as a primary oxidation product, was quite similar to the lipoxygenase activity isolated from many terres-

trial plants,⁵⁹ but different from the known 12-lipoxygenase isolated from the following anadromous fish tissues and marine algae: rainbow trout skin⁶ and gills,⁷ lake trout gills,⁷¹ two kinds of smelt skin,^{60,66} and several red algae.⁷² The sardine lipoxygenase was also different from 15-lipoxygenase, which was much postulated by the frequent occurrence of C₆-compounds in freshwater fish volatiles.^{9,13} Despite the higher contents of EPA and DHA in sardine than those in freshwater fish, the sardine lipoxygenase reacted with those PAs a little, but did well with LA (Table 6). A smelt skin lipoxygenase exhibited a good response in the PAs, but a negligible response to LA.⁶⁶ Also, the lipoxygenase of rainbow trout in the gill tissue exhibited similar reactivities toward ARA, EPA, and DHA, but low reactivity toward LA.⁷ Therefore, their contrary

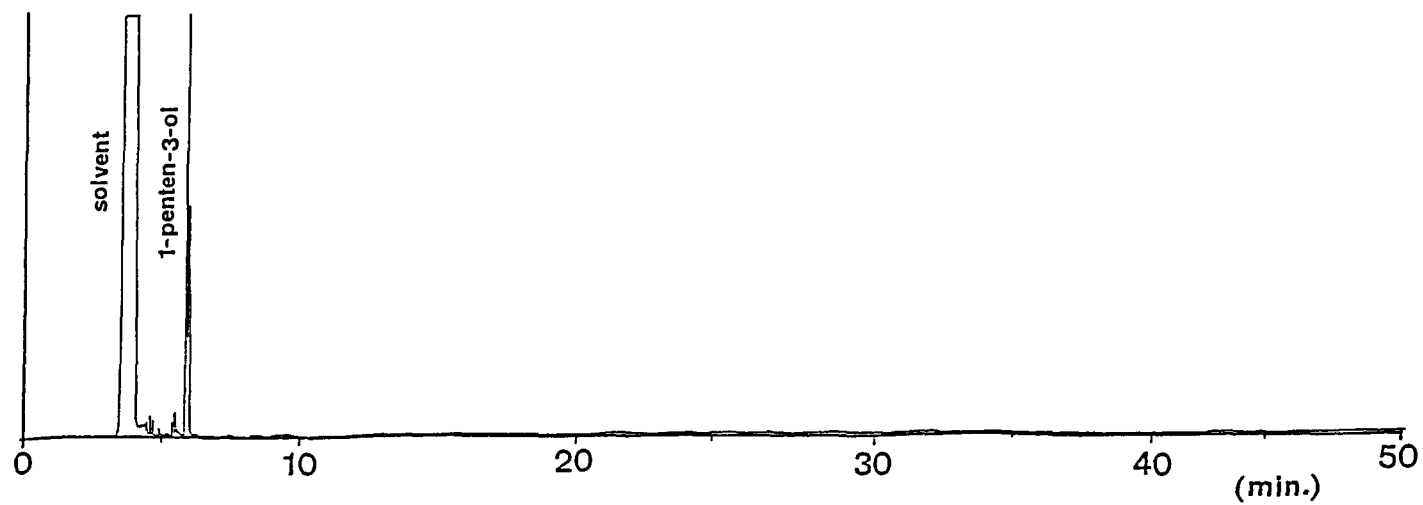


FIGURE 5. Gas chromatogram of muscle with skin of fresh sardine purchased from a local market. The vapor arising from the sardine heated in a temperature range of 22 to 99°C was trapped with dry ice/methanol and submitted to the GC (0.25 mm i.d. × 60 m nonpolar and capillary column, DB-1). The compound in the second peak remains unidentified. (From Komi et al.⁶⁸ With permission.)

reactions against those PAs seem to cause the difference in volatile constitution pattern between the empty monotony from sardine and the abundant variety from ayu smelt and rainbow trout.

The difference in enzyme activity of fresh fish was also observed, as shown in Table 7,⁷ where 12-lipoxygenase activity was species specific. There was little correlation in lipoxygenase activity between gill and skin tissues of the same fish species.⁷ However, lipoxygenases in the skin tissues of anadromous (rainbow trout and salmon) and freshwater fish (channel catfish, gizzard shad, and brown bullhead) were more reactive toward 12-hydroxy-ARA than those of ordinary euryhaline (pumpkin seed, freshwater drum, smallmouth bass, white and yellow perch, white and rock bass, and blue gill) and a saltwater fish (sheephead). Because many volatile compounds in Tables 3 to 5 were found in the extracts of fish with skins, the difference in reactivities of skin lipoxygenases to PAs must have affected that volatile constitution. The difference in volatile constitution among saltwater, freshwater, and euryhaline fish is possibly based on the difference of each enzymatic reactivity.

3. Sardine and Other Fatty Fish

No other characteristic components than 1-penten-3-ol has been detected in fresh sardine from our survey (Figure 5). However, this alcohol remains questionable as a fresh sardine volatile, because the headspace analysis on fresh and boiled sardine represented a few faint peaks on the GC charts where only propanol was identified,⁶⁷ and from the roasted sardine, the alcohol appeared at

last.⁶⁹ The monotonous GC pattern was also obtained from fresh sardine minced meat:⁷³ a total volatile amount of the sardine, together with indistinct odors, continued unchanged for the storage in some temperature ranges of 25 to 50°C for 15 min, suggesting that none of the oxidative enzymatic reaction against self-oxidation was allowed to occur. After that, the amount increased rapidly above 60°C with moderate emission of a different odor. The temperature that gave the odor alteration was almost the same as that of sardine skin lipoxygenase, at which activity was thermally deteriorated.⁶¹ Above the thermal boundary at 60°C, interactions of low bp compounds or something degraded possibly began to form the heated sardine volatiles. Therefore, 1-penten-3-ol was proposed as an enzymatic oxidation product (Figure 1), but it might have been formed by nonenzymatic oxidation in sardine.

Generally, the heating process makes hidden volatile components become visible and identifiable. Headspace components arising from several roasted salt- and freshwater fishes and other marine products are summarized in Table 8. The following characteristic components are listed:¹⁸ 2-oxolanone (γ -butyrolactone) for mucous fishes (common conger, small-mouthed sole, eel, catfish, and loach); 1-penten-3-ol for sardine; and phenethyl alcohol for Japanese common conger. The mutually common volatiles between the constitution in Table 8 from salt- and freshwater fishes and that in Table 3 from the euryhaline fish, salmon B, are few.

From the boiled sardine muscle, 40 volatile components were identified, including 14 hydrocarbons, 9 saturated aldehydes, 7 saturated alcohols, 5 amines, 2 esters, and 3 miscellaneous.⁷⁴

TABLE 6
Substrate Specificity of Partially Purified Sardine Skin Lipoxygenase

Substrate	Relative activity	Substrate	Relative activity
Linoleic acid	100	Methyl linoleate	130
α -Linolenic acid	113	Trilinolein	167
γ -Linolenic acid	38	L- α -Lecithin dilinoleoyl	30
Arachidonic acid	45	9E,12E-octadecadienoic acid	0
Eicosapentaenoic acid	22	Docosahexaenoic acid	44

From Mohri et al.⁶¹ With permission.

TABLE 7
Relative Activity (Rel. Act.) of Lipoxygenases in Gill and Skin Tissues of Fish

Fish species	Re. act. compared to trout tissue (%)	
	Gill	Skin
Freshwater drum (<i>Aplodinotus grunniens</i>)	160	17
White perch (<i>Morone americanus</i>)	156	8
Emerald shiner (<i>Notropis atherinoides</i>)	154	40
White bass (<i>Morone chrysops</i>)	151	nd
Gizzard shad (<i>Dorosoma cepedianum</i>)	140	50
Rock bass (<i>Ambloplites rupestris</i>)	136	nd
Brown bullhead (<i>Ictalurus nebulosus</i>)	133	36
Sheephead	124	nd
Rainbow trout (<i>Salmo gairdneri</i>)	100 ^a	100 ^b
Channel catfish (<i>Ictalurus punctatus</i>)	88	84
Salmon (<i>Salmo salar</i>)	79	44
Smallmouth bass (<i>Micropterus dolomieu</i>)	43	13
Yellow perch (<i>Perca flavescens</i>)	40	5
Blue gill (<i>Lepomis macrochirus</i>)	31	nd
Pumpkin seed (<i>Lepomis gibbosus</i>)	26	20

^a 100% activity corresponded to the formation of 30 nmol of 12-hydroxy-5,8,10,14-eicosatetraenoic acid by 1 mg of partially purified gill lipoxygenase from rainbow trout following a 10-min incubation.

^b 100% activity corresponded to the formation of 20 nmol of 12-hydroxy-5,8,10,14-eicosatetraenoic acid by 1 mg of partially purified gill lipoxygenase from rainbow trout following a 10-min incubation.

From Hsieh et al.⁷ With permission.

Among them, TMA and other amines were dominant. In another report,⁷⁵ volatiles arising from grilled sardine and mackerel were collected under reduced pressure. From this sardine, 8 C₁ to C₆ acids (0.4 mg/100 g sardine in total), 11 saturated, and 7 unsaturated aldehydes were identified. Because two identifiable butanoic and pentanoic acids (0.04 and 0.01 mg/100 g sardine) have low thresholds (3.0 and 1.1 ppm),²⁸ both were concluded to be the characteristic components of the grilled sardine.⁷⁵ However, because the acids have a higher bp than the amines, the acids might have been formed from decomposition of salts of amines resulting from the removal of amines under the reduced pressure.

Comparing the two reports, the former reported that sardine contained amines in large quantities but no acids, and the latter contained acids

but no amines. Simultaneous generation of amines and acids confuses us for reconstituting the flavor by using synthetic compounds, due to lack of pH values in the distillates. Because the odor quality of fish depends on pH (Table 1), information on the pH of steam distillates before the submission to GC analysis should be known.

The alteration of the volatile constitution of sardine during heating up to 230°C is shown in Table 9,⁶⁸ following Figure 5. The pH alteration of the distillates will be exhibited (as no. 2 item in Table 12). The listed compounds, obtained from a low temperature range from 99 to 101°C, included only seven common components to the salmon B volatiles in Table 3: 1-penten-3-ol (15, a number in Table 9), hexanal (27), hexanol (37), 4Z-heptenal (41), and three saturated ketones (40, 64, and 69). Among the listed compounds, indole

TABLE 8
Volatile Compounds Obtained from Head Space of Saltwater Fishes (A-E),
Freshwater Fishes (F-I), and Sea Shellfish (J-L)

Compound	Saltwater fish					Freshwater fish				Sea shellfish		
	A	B	C	D	E	F	G	H	I	J	K	L
Carbonyls												
Ethanal	+ ^a	+	+	+	+	— ^b	+	+	+	+	+	+
Propanal	—	+	—	—	—	—	—	—	—	—	—	—
2-Methylpropanal	+	+	—	—	—	—	+	+	+	—	—	+
3-Methylbutanal	+	+	+	+	+	+	+	+	+	+	+	+
Acetone	+	—	—	+	—	—	—	+	+	—	—	+
2-Butanone	—	—	—	—	—	—	+	—	—	—	—	—
Benzaldehyde	+	—	+	+	+	—	+	+	—	+	—	+
2-Phenylethanal	+	+	+	+	+	+	—	+	+	+	+	+
Alcohols												
Ethanol	—	+	—	—	—	—	—	—	—	—	—	—
1-Penten-3-ol	—	+	—	—	—	—	—	—	—	—	—	—
Furfuryl alcohol	+	+	+	—	—	+	—	+	+	+	—	+
Phenethyl alcohol	—	—	+	—	—	—	—	—	—	—	—	—
N-compounds												
Pyridine	—	—	—	—	—	—	—	—	—	+	—	+
2-Methylpyridine	—	—	—	—	—	—	—	—	—	—	—	+
3-Methylpyridine	—	—	—	—	—	—	—	—	—	—	—	+
2-Methylpyrazine	—	—	—	—	—	—	—	—	—	—	—	+
2,5-Dimethylpyrazine	+	+	—	—	—	—	+	+	+	—	—	—
2,6-Dimethylpyrazine	—	—	—	—	—	—	—	—	—	—	—	+
2,3,5-Trimethylpyrazine	+	+	+	+	+	+	+	+	+	+	+	+
Hydrocarbons												
Decane	—	—	—	—	—	+	—	—	—	—	—	—
Undecane	—	—	—	—	—	+	—	—	—	+	—	—
Tetradecane	—	—	—	—	—	—	—	—	+	—	—	—
Pentadecane	+	+	+	—	—	+	—	—	+	—	—	—
Hexadecane	—	+	—	—	—	—	—	—	+	—	—	—
Heptadecane	+	+	+	—	+	+	+	+	+	—	—	—
1-Heptadecene	—	+	—	—	—	—	—	—	+	—	—	—
Octadecane	—	—	—	—	+	—	—	—	+	—	—	—
1-Octadecene	—	—	—	—	—	—	—	—	+	—	—	—
Nonadecane	—	—	—	—	—	—	—	—	+	—	—	—
2,6,10,14-tetra-methylpentadecane	+	+	+	—	+	+	+	—	—	—	—	—
Miscellaneous												
Dimethyl sulfide	—	—	—	—	—	—	—	—	+	—	—	—
Toluene	+	+	—	—	—	+	—	+	+	—	—	+
2-Oxolanone	—	—	+	+	—	+	+	+	—	—	—	—

Note: A, Yellow tail (*Seriola quinqueradiata*); B, sardine (*Sardinops melanostictus*); C, common Japanese conger (*Conger myriaster*); D, smallmouthed sole (*Limanda herzensteini*); E, Japanese whiting (*Shillago japonica*); F, eel (*Anguilla japonica*); G, Japanese catfish (*Silurus asotus*); H, loach (*Misgurnus anguillicaudatus*); I, common minnow (*Zacco platypus*); J, Ocellate octopus (*Octopus ocellatus*); K, oval squid (*Sepioteuthis lessoniana*); L, Joyner's shrimp (*Metapenaeus joyneri*).

^a Identified.

^b Not detected.

From Kasahara.¹⁸ With permission.

TABLE 9

Volatile Compounds Identified in Sardine from Five Continuous Temperature Ranges (99–101, 101–115, 115–160, 160–200, and 200–230°C)

No.	Compound	Sardine (ppm)				KI (DB-1)		
		99–101	101–115	115–160	160–200	200–230	Unknown	Known ^a
1	Ethanal	0.07	0.19	0.37	0.55	1.57	442	453
2	Methanethiol	0.07	0.08	0.36	0.54	1.30	453	470
3	Propanal	0.07	0.08	0.22	0.52	4.05	481	491
4	Diethyl ether (solvent)						482	477
5	Dichloromethane (solvent)						510	511
6	2-Methylpropanal	0.12	0.51	1.36	0.60		520	515
7	2-Butanone	0.07	0.09	1.69	4.29	17.13	557	560
8	2-Methylpropanenitrile			0.06	0.18	0.84	580	575
9	Ethyl acetate	0.03	0.02	0.15	0.33	0.45	594	590
10	3-Methylbutanal	0.26	0.31	0.43	0.65	2.21	622	626
11	Butanenitrile		0.02	0.24	0.19	0.56	627	
12	2-Methylbutanal	0.20	0.42	0.95	0.40	1.17	636	637
13	2-Pentanone			0.06	0.11	0.26	650	646
14	Propyl isocyanate			0.20	0.05	0.22	653	650
15	1-Penten-3-ol	0.32					656	656
16	3-Hydroxy-2-butanone	0.08					676	673
17	3-Methylbutanenitrile	0.03	0.11	0.90	2.12	1.64	687	685
18	2-Methylaminoethanol			0.17	2.04	14.36	706	708
19	Methyl lactate	0.05	0.17	0.53	1.01	1.34	709	710
20	1-Methylpyrrole			0.09	0.20	0.49	713	717
21	Pyridine and pyrrole	0.14	0.31	1.81	3.72	7.54	715	— ^b
22	Dimethyl sulfide		0.22	0.06	0.07		728	728
23	<i>N,N</i> -dimethylformamide		0.02	0.04	0.21	0.21	732	735
24	2 <i>Z</i> -pentene-1-ol	0.07					742	742
25	Toluene	0.13	0.08	0.60	1.28	6.82	752	752
26	Cyclopentanone	0.11	0.32	0.48	0.40		760	763
27	Hexanal	0.04					777	775
28	<i>N</i> -ethylidene-2-methyl butylamine		0.02	0.04	0.10	0.70	788	790
29	Ethyl lactate	0.04	0.03	0.15			789	790
30	2-Methylpyridine		0.07	0.07	0.40		799	798
31	2-Methylpyrazine	0.01	0.06	0.07	0.14	0.26	803	804
32	2-Methylpyrrole		0.07	0.41	0.87	1.59	811	812

TABLE 9 (continued)

Volatile Compounds Identified in Sardine from Five Continuous Temperature Ranges (99–101, 101–115, 115–160, 160–200, and 200–230°C)

No.	Compound	Sardine (ppm)				KI (DB-1)		
		99–101	101–115	115–160	160–200	200–230	Unknown	Known ^a
33	1-(methylthio)-1-ethanethiol		0.08	0.13			823	
34	Furfuryl alcohol	0.29	0.36	1.01	1.41	10.36	825	830
35	<i>N,N</i> -Dimethylacetamide			0.24	0.83		831	834
36	3-Methylpyridine	0.01	0.07	0.13	0.40	1.04	831	834
37	Hexanol	0.08					849	849
38	Ethylbenzene			0.15	0.16	0.45	846	850
39	2-Oxolanone	0.03	0.06	0.27	0.48	0.56	859	859
40	2-Heptanone			0.06			863	867
41	4 <i>Z</i> -heptenal	0.04					878	874
42	Styrene			0.06	0.12	0.57	878	880
43	2-Acetylfuran		0.03	0.08	0.11	0.27	882	883
44	2,5-Dimethylpyrazine	0.12	0.15	0.71	1.14	1.24	881	883
45	2-Ethylpyrazine				0.07	0.35	890	893
46	Nonane	0.03					900	900
47	2,5-Dimethylpyrrole				0.88	1.79	904	906
48	2,4-Dimethylpyridine			0.09	0.28	0.85	906	908
49	<i>N,N</i> -Dimethylpropionamide			0.05	0.38	0.96	917	918
50	2,3-Dimethylpyridine		0.04	0.09	0.12	0.62	923	920
51	<i>N</i> -(2-methylpropylidene)-3-methylbutylamine			0.14	0.85	1.82	926	927
52	3-Ethylpyridine			0.28	0.49	0.71	932	932
53	Benzaldehyde	0.23	0.04				935	935
54	Propylbenzene				0.46	0.55	940	940
55	Dimethyl trisulfide	0.02					951	951
56	2-Methylbutyl propanoate	0.11					954	953
57	Phenol	0.08	0.50	2.33	2.21	6.57	958	956
58	2,3,5-Trimethylpyrazine	0.02	0.07	0.37	1.14	1.49	982	983
59	Phenylacetaldehyde	0.13	0.19				1016	1017
60	<i>N</i> -(2-methylbutylidene)-3-methylbutylamine			0.25	0.27	0.87	1028	1026
61	2-Acetylpyrrole	0.13	0.02	0.37	0.79		1036	1037
62	1-Methylpyrrolidine-2,5-dione	0.02	0.16	0.53	1.29	4.23	1038	1042

63	4-Methylphenol and/or 3-methylphenol	0.08	0.25	1.61	4.96	3.97	1052	— ^c
64	2-Nonanone	0.08	0.04				1073	1070
65	Phenylacetonitrile	0.03	0.09	0.47	0.36	1.47	1093	1094
66	Undecane (an internal standard)						1100	1100
67	1-Acetylpyrrolidine		0.02	0.89	0.97	1.81	1127	1128
68	Dimethylphenol			0.24	0.25	0.51	1148	
69	2-Undecanone	0.06	0.03	0.13			1174	1172
70	5,6-Dihydro-2,4,6-tri methyl-4 <i>H</i> -1,3,5-dithiazine		0.03	0.11			1181	1181
71	Benzothiazole	0.03	0.04				1196	1196
72	4-Phenyl-2-butanone				0.13	2.28	1210	1210
73	Indole	0.49	1.68	5.65	9.66	16.72	1264	1264
74	2-Dodecanone	0.05					1275	1276
75	3-Methylindole	0.11	0.22	0.64	0.73	1.74	1360	1360
76	Tetradecane	0.02					1402	1400
77	2,6-Di- <i>tert</i> -butyl-4- methylphenol (BHT)	0.08	0.32	0.46	0.49	0.46	1497	1495
78	Pentadecane	1.50	0.27	0.69	0.90	1.68	1502	1500
79	Hexadecane	0.22	0.09				1602	1600
80	Heptadecane	2.36	0.57	1.17	1.21	0.76	1702	1700
81	Octadecane	0.15					1801	1800
82	Proline anhydride	0.06	1.99	1.05	1.42	3.97	1867	1870
83	Nonadecane	0.04					1898	1900
84	Dibutyl phthalate	0.08	0.09				1903	1906
85	Methyl hexadecanoate	0.04	0.12	0.22	0.24	0.50	1905	1907
86	Tetradecanamide	0.09	0.05	0.12	0.33	0.58	1933	
87	Methyl octadecanoate				0.08	0.32	2107	2109
88	Hexadecanamide	0.37	0.45	0.96	1.86	5.61	2140	2142
89	<i>N</i> -methylhexadecanamide	0.02	0.04	0.15	0.25	0.66	2175	2178
90	Octadecanamide	0.05	0.16	0.22	0.18	0.21	2341	2341
	Total	9.26	11.52	33.63	58.87	143.26		

^a Compounds accompanying no KI (Kovats Index) data are tentative identification, solely based on mass spectrum matching.

^b KI values of authentic pyridine and pyrrole are 712 and 727.

^c KI of authentic 3- and 4-methylphenols are 1047 and 1050.

From Komi et al.⁶⁸ With permission.

(73, 0.49 ppm), 1-penten-3-ol (0.32 ppm), 3-methylindole (75, 0.11 ppm), and 4Z-heptenal (0.04 ppm) can contribute to the odor of sardine. The large gap in volatile constitution between Tables 3 and 9 suggested that formation mechanisms for volatiles between sardine and salmon B are essentially different from each other. The gap may depend on an antioxidation system: a holder (sardine) and a dropper (salmon B).

4. Cod and Other Lean Fish

Like sardine, fresh cod has an indistinct odor.⁴⁵ It gives off many aldehydes as follows in order of quantity; hexanal, heptanal, 2E,4Z-heptadienal, 2E-hexenal, 4Z-heptenal, and 2E,4Z-decadienal.⁷⁶ These components occurred in very small amounts in the fresh cod fillet: for example, hexanal, 2 ppb (T, 4.5 ppb in Table 5); 4Z-heptenal, 0.2 ppb (T, 0.04 ppb).⁷⁶ The aldehyde, 4Z-heptenal, due to its low threshold value in water and the odor evaluation, was concluded to be the characteristic component for off-flavor of cold-stored and raw cod among the 95 identified compounds. This aldehyde is a common flavor component in salmon B (Table 3) and sardine (Table 9). However, the aldehyde in the cod, as well as other unsaturated aldehydes, might have been formed by the thermal effect of steam distillation applied for isolation of the volatiles. From frozen Pacific halibut stored at -20°C, ethanal was identified as the main carbonyl compound, together with small amounts of methanal, nonanal, and 4 ketones (C₃ to C₆), by using two methods of reduced pressure steam distillation and hexane extraction.⁷⁷ Also, from nonirradiated haddock flesh, only a simple volatile constitution (14 saturated compounds in total) was reported.⁷⁸ Our recent study on volatiles of fresh cod muscle with skin gave no peaks to the GC chart in a temperature range of 22 to 99°C. From the distillate at a range of 99 to 101°C, 7Z,10Z-hexadecadienal (0.02 ppm) was the characteristic carbonyl compound, along with a major component, indole (0.58 ppm). The aldehyde was the only unsaturated aldehyde among 62 components identified after heating the cod to 240°C. It has a powerful, pleasant marine odor that resembles that of 5Z,8Z,11Z-tetradecatrien-2-one.¹⁹ However, for the heated cod odor, indole can more affect the odor quality than the aldehyde.

As described above, slight detection of mutually common unsaturated carbonyl and alcohol volatiles between saltwater and euryhaline fish suggests that the compounds listed in Table 5 mostly belong to the latter. Saltwater fish seem to prevent from oxidizing itself by using an unknown system, resulting in almost no production of any carbonyl and alcohol compounds derived from PAs through enzymatic or nonenzymatic biooxidation despite fatty and lean properties.

E. N-Compounds and Low Fatty Acids

Contrary to slight detection of carbonyl and alcohol compounds in live saltwater fish, non-volatile trimethylamine *N*-oxide (TMAO), the precursor of TMA, has been found in abundance.^{4,79,80} TMAO in live saltwater fish was little changed, but it degraded into TMA during storage and thermal processing.⁸⁰ TMAO and TMA were more abundant in saltwater fish than in freshwater fish as exemplified as follows:⁸⁰ TMAO, 1000 mg/100 g pollack muscle, and TMA, 3 mg; TMAO, 2 mg/100 g common minnow muscle, and TMA, nil. These compounds are characteristic of saltwater fish, because of the rare detection of TMA in other foods as mentioned above. Besides, its relatively low odor threshold value (600 ppb) among those compounds in Table 2 can contribute to the saltwater fish odor.

Of the volatile bases listed in Table 2, the odor of ammonia is strongest, followed by that of TMA and DMA.⁴ Ammonia occurred at 24 mg/100 g albacore muscle (240 ppm in the muscle) in thawed albacore, this amount being 40 and 120 times as much as TMA and DMA, respectively.⁸¹ These amounts (240, 6, and 2 ppm) are 2, 10, and 0.1 times larger than each odor threshold value in water (Table 2). According to the comparison, TMA might have contributed to the albacore odor. However, a fishy odor is rarely found in albacore, even when amounts of TMA reached 30 to 200 times the threshold value (0.6 ppm).⁸¹ This phenomenon was probably caused by the coexistence of acids as shown in Table 1.⁴

The odor of ammonia, despite its high odor threshold (110 ppm), has often been smelled from a sealed pack of chilled flounder fillet, the result of the accumulation of ammonia in the pack. More significant than the odor problem, ammonia pro-

vides a basic environment that accelerates generation of other compounds, such as aldehydes (Figure 4) and H_2S (Figure 6). This results in the formation of flavor compounds, for example, ammonia increases formation of pyrazines⁸² and other N-compounds that react with H_2S to form *S,N*-heterocycles.^{42,43}

To suppress the generation of ammonia and TMA, CO_2 and some nonvolatile acids have been added to fish.⁸³ To remove TMA, the addition of vitamin U (3-amino-3-carboxypropyl dimethyl sulfonium salt) has been tested.⁸⁴ Bacterial activity on fresh fish odors seems negligible under cold-storage conditions. Incidentally, ammonia is generated not only from fish, but also from other meats. For example, ammonia accounted for 99% of the total bases found in beef.⁸⁵

On the other hand, formic, acetic, and propanoic acids were found, together with TMA almost simultaneously, in small amounts in fresh saltwater fishes.^{28,86} For example, these acids (10, 1, and 0.4 mg/100 g muscle, respectively) were determined in fresh herring muscle.⁸⁶ They were derived from the salts of TMA.³ Despite their high odor thresholds (1000, 34.2, and 32.8 ppm in Table 2), the odor arising from each mixture of TMA with the acid changed greatly with a change in pH, the acid concentration, as shown in Table 1. With an increase in the acid, TMA altered its odor quality, decreasing its volatility. At nearly neutral pH, the mixture emitted a boiled or canned fish odor. TMA has a lower odor threshold than each acid, but the change in odor quality was evidently influenced by the combination.²⁸

The pH effect on the odor of a euryhaline fish was reported by Josephson et al.⁸⁷ A fresh whitefish (a trout) extract lost its characteristic odor at high pH, and addition of sodium bisulfite completely suppressed its fishy odor in a range of pH 5.5 to 8.5, as can be seen in Table 10.⁸⁷ The GC-MS analysis of the extract treated with bisulfite at pH 7 detected a few identified compounds, whereas 63 compounds were found in the untreated extract.⁸⁷ Addition of dilute solutions of HCl and NaCl, instead of bisulfite, also changed the odor quality (Table 10). Thus, pH greatly influenced the volatile constitution and the odor quality. The aroma properties listed in Table 10 suggest that pH dominates fish volatiles even though lipoxygenase activity increases or de-

creases. Some lipoxygenase activities are confined to a pH range as follows: from rainbow trout gills, the pH range was 7 to 9 with optimum pH at 7.5;⁷ from ayu smelt skin, almost the same range with the optimum at 7.4;⁶⁰ from the sardine skin, the range 6 to 9 with the optimum 7.0.⁶¹ In addition to the enzymatic activity, pH affects retroaldol condensation of aldehydes (Figure 4). For example, formation of 4Z-heptenal is greatly accelerated by higher pH (pH 11) at even low temperature (21°C). Thus, pH has a close relationship with odor quality and volatile constitution.

Information concerning pH relating to odor quality is useful in comprehending the compounds appearing in GC peaks, and reconstituting the fish flavor by using synthetic compounds based on the GC data. Practically, some combinations of TMAO and acids at a definite pH level have been used for manufacturing flavors. However, no combinations of TMA with any carbonyl or alcohol compound have been reported.

From the physiological point of view, odorless saltwater fish are possibly influenced by the high salinity (about 3.5%) and the alkalinity (about pH 8.2) of the sea water. Against the environments, TMAO occurs abundantly in saltwater fish, but seldom in freshwater fish.^{88,89} TMA played an important role in keeping osmoregulation⁸⁰ and pH stable,⁴ just as glycine betaine did in marine mollusks and crustaceans.⁸⁰ An increase in TMAO in salmon and eel was reported when both fishes were transferred to salt water from fresh water.⁹⁰ However, any suppressive effects of TMAO on formation of carbonyl and alcohol compounds remain unsolved.

F. S-Compounds

S-compounds do not contribute significantly to fish odor as reported,⁴⁵ based on the detection of only three S-compounds (DMS, carbon disulfide, and H_2S) from cod just after thawing. From fresh mackerel placed at 35°C for 10 h, small amounts of H_2S and methanethiol (5 and 2 µg/100 g muscle, respectively) were barely determined. Simultaneously, a large quantity of N-compounds (14 mg/100 g in total) consisting mainly of ammonia was measured.⁹¹ It is remarkable that the generation of ammonia preceded that of H_2S . It concerns formation mechanisms for

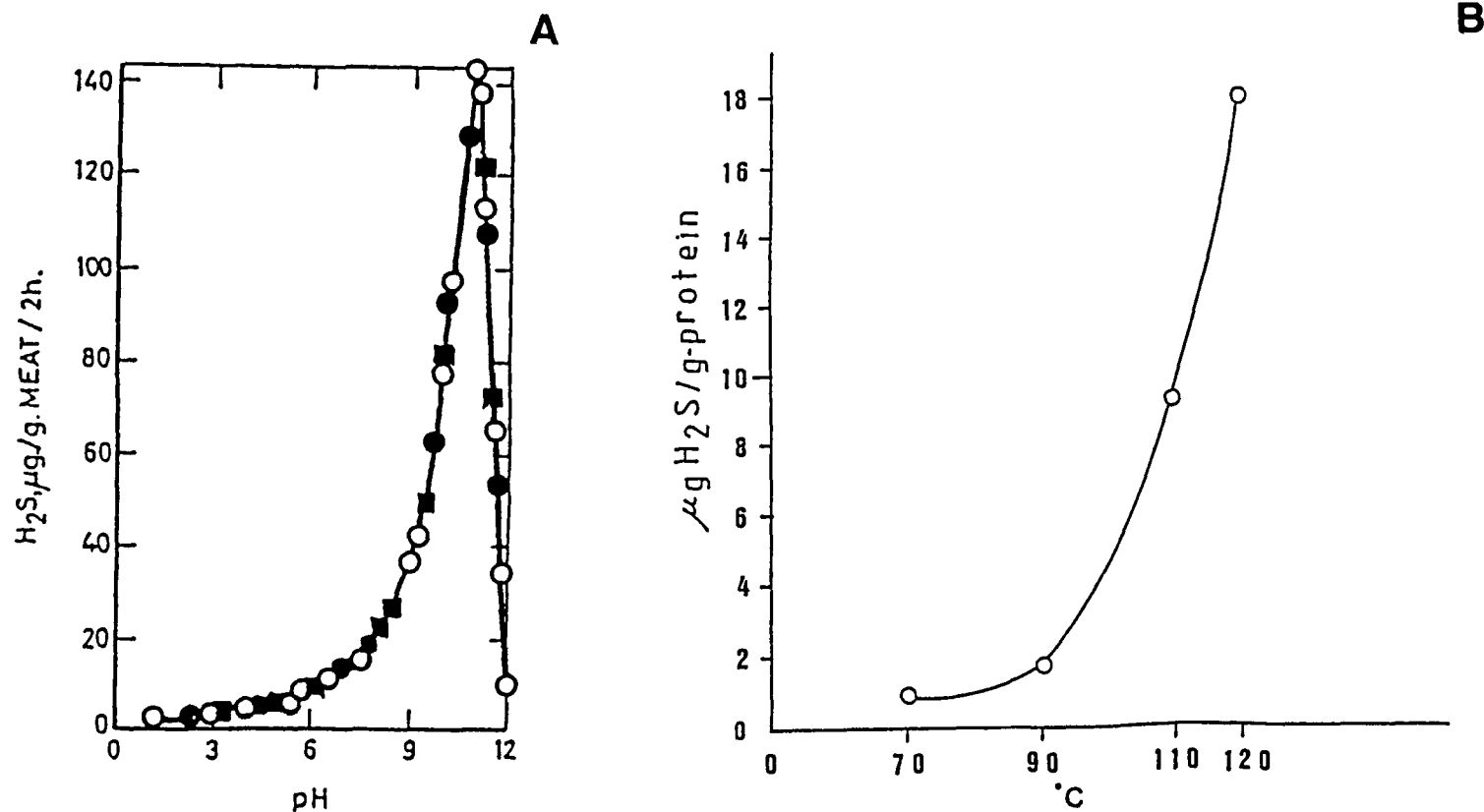


FIGURE 6. Effects of pH and temperature on H₂S formation. (A) Effects of pH on H₂S formation from some boiled domestic meats (○, mutton; ●, beef; ■, pork). (From Johnson and Vickery.²⁹ With permission.) (B) Effects of heating up to 120°C (30 min) on H₂S formation from myofibrils. (From Hamm and Hofmann.³⁰ With permission.)

TABLE 10
Effect of pH and Bisulfite (500 ppm) on the Aroma Property of Fresh Whitefish Water-Slime Extracts

pH ^a	Without bisulfite		With bisulfite	
	Odor quality	Odor intensity ^b	Odor quality	Odor intensity
2.8	Mild green, sl. cucumber	++	Sharp bit of SO ₂ , lacks fishiness, SO ₂ dominates	++++
3.8	Harsh-green, vine-like cucumber, sl. rhubarb	+++	Scalded, picked fowl, raw mammal flesh, seaweed background	+++
4.8	Strong 2,6-nonadienal	+++	Musty, hint of putrid, marine bayshore-like	++
5.8	Fresh, sweet, cucumber	+++	Bayshore, musty seaweedy, overripened melon	++
6.8	Fresh, melon, green, fishy	++++	Raw meat, blood-like, lacks fishiness	+
7.8	Distinct papery, lacks freshness, suppressed cucumber	+++	Ripe grapefruit, melony fresh, not fishy	+
8.8	Like pH 6.8 but lacks fishiness	+++	v. sl. cooking crab pot, sweet, sl. melon, sl. grapefruit	++
9.8	Same as pH 8.8	+++	Distinct cooked crab-like, ripe grapefruit	++
10.8	Sl. frozen cooked crab in shell, old whitefish	++++	Strong old fish oil, crab-like, cured salmon eggs	+++
11.8	Cured salmon eggs, frozen cooked crab in shell	++++	Sl. cured salmon eggs, frozen whole cooked crab	+++

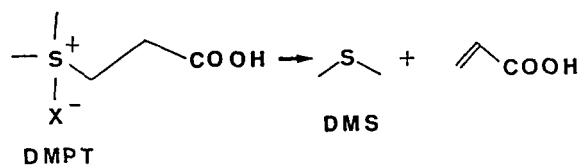
^a At ca 21°C.

^b Scale: 1 = very weak to 10 = very strong; + = weak (1–3), ++ = distinct (3.5–5), +++ = strong (5.5–7), ++++ = very pronounced (7.5–10).

From Josephson et al.⁸⁷ With permission.

flavor compounds, especially *N*-, *S*-, and *S,N*-heterocycles as will be shown.

Among the *S*-compounds in Table 2, DMS most expresses the odor giving many images of fresh seashore-animals and seaweeds. It is derived from a precursor, DMPT, which is decomposed with great ease into DMS and 2-propenoic acid (acrylic acid) by three processes: enzymatic, alkaline, and thermal decomposition.³²



SCHEME 2. Decomposition of DMPT to DMS and 2-propenoic acid.

Because microalgae, planktons, and seaweeds contain DMPT in large quantities, the presence of DMS in shellfish and some other types of fish is the result of DMPT in the food chain.^{31,32} Due to the ease of decomposition, DMPT is a unique precursor of DMS for sea creatures, compared with more stable precursors of DMS for terrestrial plants: *S*-methyl-L-cysteine sulfoxide for allium plants⁹² and vitamin U for cabbage and tea.⁹³ The physiological roles of DMPT and DMS for those creatures are not well understood.

Although DMS is suspected to cause offensive odors of fresh fish and other sea creatures,²⁵ it has been used as a flavoring ingredient for seafoods, so that some combinations of DMS and contaminants should be ascribable to the off-flavor. A powerful off-flavor substance (T, 0.01 ppb

in bland food) was reported. For 4-mercapto-4-methyl-2-pentanone having a catty odor was almost spontaneously formed from reaction of H_2S with 4-methyl-3-penten-2-one used for a chemical resin.⁵

Generation of H_2S from muscle is mainly influenced by pH and temperature.^{29,30} H_2S arising from some boiled domestic meats has the same patterns regardless of the kind of meat, and it increased most at nearly pH 11 (see Figure 6A).²⁹ Also, H_2S from a protein, myofibrils, increased greater at higher temperatures (Figure 6B).³⁰ The formation of H_2S from surimi behaved the same

as that from the protein.⁴ Moreover, H_2S from dried squid heated up to 230°C was described to have been multiplied by pH and temperature.⁴³ The production of H_2S reached 750 mg/100 g dried squid in total as presented in Figure 7,⁴³ where the neutral pH (6.8) of the mixture of dried squid with water before heating became greater (pH 9.8) just after the beginning of heating. It indicated that some basic compound with low bp (probably ammonia) occurred in this initial stage. In addition to that, the recovery of components is shown in Table 11,⁴³ where a greater amount of nitrogen (1.8 g/100 g dried squid) released than

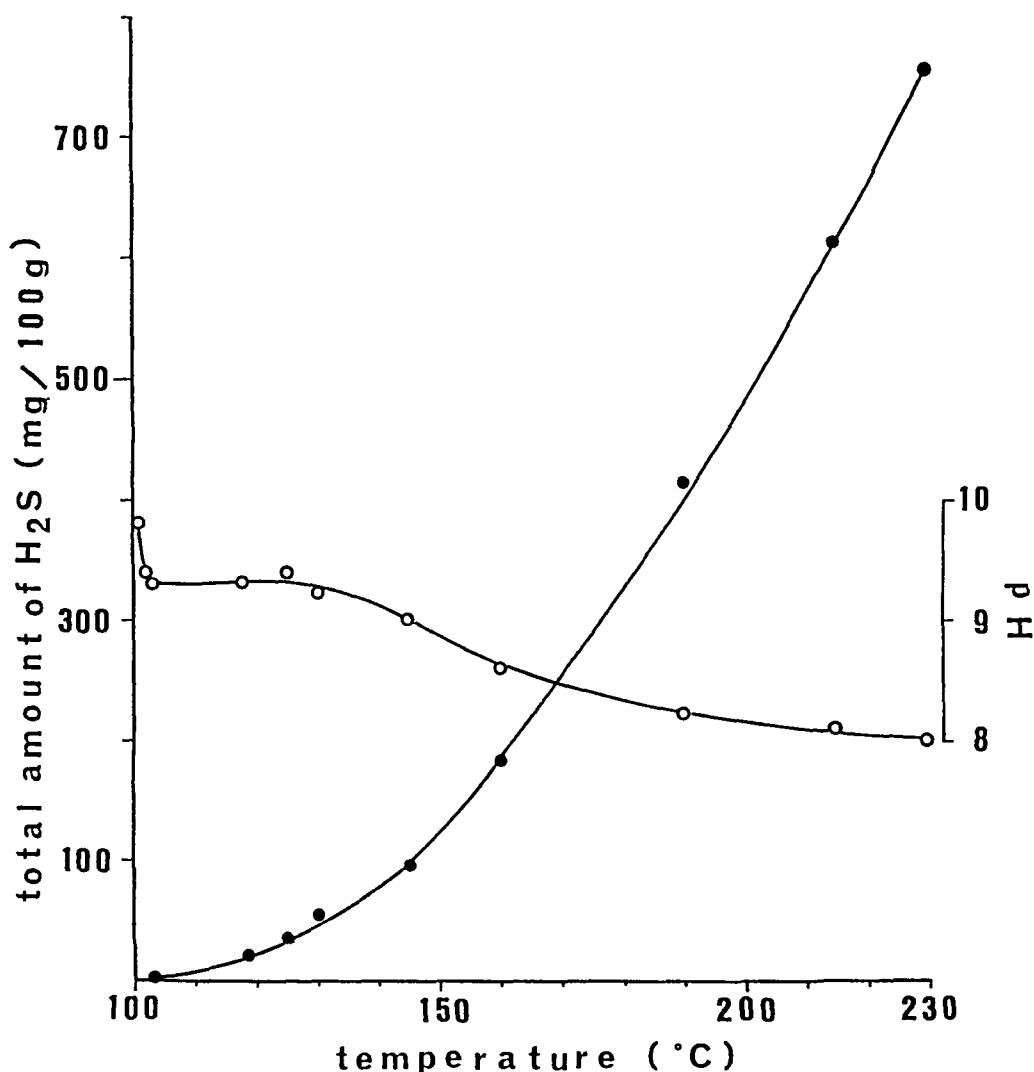


FIGURE 7. Thermal effects on H_2S formation and pH alteration after heating dried squid to 230°C (●, H_2S ; ○, pH). (From Kawai.⁴³ With permission.)

TABLE 11
Recovery of Components (g) from 100 g of Dried Squid

	Heating		Loss on heating	Gain from distillates	Balance
	Before	After ^a			
H ₂ O ^b	123.0 ^c	1.6	121.4	129	+7
N	9.75	7.93	1.82	0.1 ^d	-1.7
S ^e	1.70	0.8-0.9	0.8-0.9	0.7	-0.1-0.2
Fat	4.00	3.91	0.09		
Ash	8.00	4.73	3.27		
NaCl	3.80	1.83	1.97		

^a Thermal residue, 59 g.

^b Loss on drying at 105°C for 4 h.

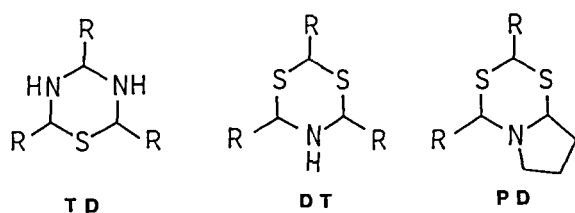
^c Moisture of the sample and water added to the system.

^d Calculated amount from (NH₄)₂CO₃ isolated.

^e Oxygen-flask method.

From Kawai.⁴³ With permission.

sulfur (0.8 g/100 g dried squid) must have made the thermal environment basic. Thus, ammonia occurs prior to H₂S in the same manner of mackerel⁹¹ as described above. A combination of ammonia, H₂S, and aldehydes produces powerful flavor compounds: 3,4,5,6-tetrahydro-2,4,6-trialkyl-1,3,5-thiadiazines (TD),^{42,94} and 5,6-dihydro-2,4,6-trialkyl-4H-1,3,5-dithiazines (DT).^{43,95,96} Further, the combination plus 1-pyrroline gives pyrrolidino[1,2-*e*]-2,4-dialkyl-4H-1,3,5-dithiazines (PD).^{96,97}



SCHEME 3. TD, DT, and PD.

The effects of pH on change in odor quality of some TDs and DTs are shown in Table 12.⁹⁴ The change in odor largely occurred near equivalent points (Ep): an odor of methyl TD (3,4,5,6-tetrahydro-2,4,6-trimethyl-2H-1,3,5-thiadiazine) before the addition of diluted HCl gave strong and roasted cereal odors. As pH gradually approached the Ep value (pH 4.0), the odor altered to strong roasted popcorn odors. Thus, even a

single compound also gets an effect of pH. Incidentally, such odor change is observed simply: when a little bit of methyl or ethyl TD is put on a fingertip, the odor alters in a moment. This is probably caused by the stable pH of human skin at 5.0.⁹⁸

III. THERMAL ENVIRONMENTS OF HEATED FISH

A. Heating Method and Procedures

Favorable heated-food odors consist of boiled- and roasted-food odors simultaneously, if cooking odors can be roughly divided into these odors. The roasted-food odor has frequently been required for flavoring, due to its more pleasant, perceptible, and intense odor properties than the boiled odor. However, its drawback is unstable thermal generation. The problem has been solved by using glass beads (GB), as illustrated in Figure 8.⁹⁹

A pan for Chinese cooking has a large surface giving off an appetizing cooking odor with great moisture from small amounts of cooking ingredients. Also, a huge pan has been used with heated sands or small stones for manufacturing many roasted foods, such as chestnuts, sweet potatoes, and barley. This means that a large surface area plays a more important role than the cooking

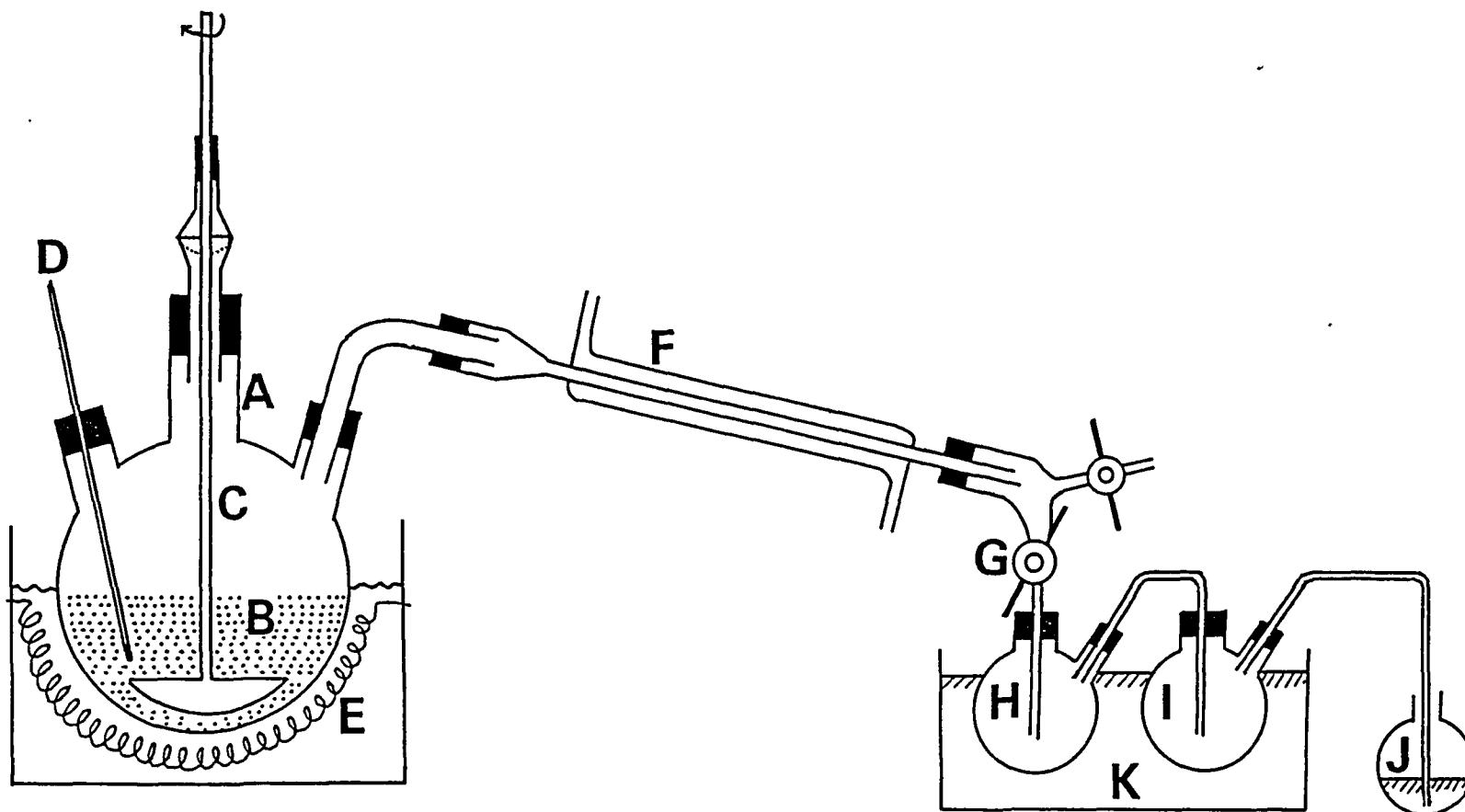


FIGURE 8. Heating apparatus: A, flask; B, glass beads; C, propeller; D, thermometer; E, mantle heater; F, condenser; G, cock; H, flask for trap; I, empty flask to prevent contents of flasks H and J from mixing with each other. J, flask with water for absorption of offensive gases. K, salt with ice or dry ice with methanol. (From Kawai and Ishida.⁹⁹ With permission.)

temperature in producing heated-food odors. To prepare our experiment, a cover was placed over a pan and the moisture collected as a distillate for GC analysis. The device, using numerous GB instead of sand or stones in a closed system, has a large effect on the stable generation of heated-food odors. Recently, the efficiency of GB has been demonstrated in aromatic rice for its quantification of 2-acetyl-1-pyrroline, a popcorn factor,¹⁰⁰ and for the determination of total mercury in foods.¹⁰¹

The GB method has the following properties: (1) 1 kg of GB (2.3-mm diameter), for example, makes a thermoconduction surface increase 30 times as much area as that of a 2-l round-bottom flask, and the large area makes the heated-food volatiles distill easily; (2) the method can continuously produce from only one sample its qualitatively different odors, raw, boiled, roasted, and charred odors, one after another, as the temperature elevates. Because the sample does not have to be changed, it is suitable for the kinetic study of changing odors. Each fraction can be submitted to GC analysis at the same time for determination of pH, odor, and other components such as H_2S ; finally, (3) the method can produce natural cooking flavors from mixtures of complex ingredients like sukiyaki (no. 35 in Table 12).

The thermal environments of various foods were studied by the GB method:^{44,102} 38 items of food were stirred with GB, and heated to nearly 230°C within 2 h by an electric mantle heater to isolate the heated-food volatiles. The distillates, fractionated by four or five temperature ranges according to the temperature of the beads, were submitted to pH measurement and odor evaluation. Some series of fractionated distillates were further analyzed by GC and GC-MS.

B. Alteration in Odors of Distillates during Heating

Odor evaluation of the distillates arising from 15 animal- and 17 plant-origin foods was performed by some flavorists. The open discussion is summarized in Table 13,^{44,102} where change in odors of distillates arising from the tested foods during heating was described. The animal-origin

food did not give off any typical Maillard reaction odors derived from amino acids with sugars. The lack of such odors, together with small contents of carbohydrates as shown in Table 12,^{44,102} has thrown doubt on the possibility of great occurrence of the typical Maillard reactions in the heated animal-origin foods. As the temperature was elevated, an $(NH_4)_2CO_3$ -like irritant odor increased characteristically, resulting from the generation of ammonia and CO_2 . In contrast, almost all the plant-origin foods produced intimate Maillard reaction odors, including burned sugar odors. Favored fractions of each sample were asterisked as those in Table 12, indicating the suitable temperature ranges for cooking.

C. Alteration in pH of Distillates during Heating

In Table 12, the principal contents (proteins, carbohydrates, and lipids) of 38 items of food are listed, as are the pH alteration of distillates arising from the samples after heating to nearly 230°C, another pH alteration of the samples before and after heating, and the detection of *S,N*-heterocycles in the fractions with the total amounts. The samples were divided into four groups according to their origin: animal, plant, both, and other. They were placed in order of high protein content in each group. Table 12 shows that (1) animal-origin foods produce basic distillates throughout heating, whereas almost all the plant-origin foods yield acidic distillates; (2) the basic (high) pH of the distillate largely depends on a high protein content, whereas the acidic (low) pH depends on a high carbohydrate content; however, both pHs do not depend on a lipid content; and (3) pHs of food are quite different from those of distillates, representing unchangeable and neutral values in foods before and even after heating.

It is remarkable that every proteinous food (e.g., wheat gluten, no. 16 in Table 12) continue to produce basic distillates during the heating process. All the seafoods, domestic, and poultry meats, except short-necked clam (13) and its dried product (5), gave basic distillates in a pH range of 9 to 10, despite no pH alteration in every thermal residue. Whereas, the plant-origin foods, except

TABLE 12

Thermal Environments of Food: The Principal Contents, Change in pH of Distillates from Heating Food in Five Temperature Ranges (fr. 1, 99–101; fr. 2, 101–115; fr. 3, 115–160; fr. 4, 160–200; and fr. 5 > 200°C), Another Change in pH of Food Before and After Heating, and *S,N*-Heterocycles in the Distillates

Food no.	Food	Sample (S)		Contents (g) of S ^a				pH					pH (sample)			Recovery			S,N-hetero cycles (ppm)
		Amount (g)	Water added (g) ^b	Protein (P)	Carbohydrate (C)	Lipid (L)	P/P+C (%)	Distillate (D) fractions					Heating		Maximum temp (°C)	Total D (g)	Residue		
								1	2	3	4	5	Before ^c	After ^d			(R) (g)	D+R/S (%)	
Animal-origin food																			
1	Dried skin ^e (s), flying squid	120	120	82	t ^f	12	99	9.6	9.4 ^g	9.3 ^g	9.1	—	6.5	5.9	204	146	88	98	+
2	Fresh sardine, muscle(m)+s	400	0	80	t	10	99	9.0	9.6	9.7 ^g	9.6	9.7	6.2	7.2	230	306	89	99	0.1
3	Dried common squid	120	120	73	t	5	99	9.8	9.3 ^g	9.1 ^g	8.2	8.0	6.8	6.6	230	155	71	94	+++
4	Fresh cod, m + s	400	0	(63)	(0)	(2)	(100)	10.3	10.0 ^g	9.9 ^g	9.8	9.9	7.5	8.2	240	325	59	96	—
5	Dried short-necked clam ^e	90	90	54	2	4	96	8.6	7.1	7.3 ^g	8.7	8.9	6.3	5.8	230	108	62	95	20.9
6	Prawn, black tiger, tail	240	0	(49)	(0)	(2)	(100)	8.5	9.0 ^g	9.1 ^g	8.9	9.1	7.2	7.9	216	183	51	97	
7	Roast pork, a manufactured product	240	0	(47)	(15)	(20)	(79)	8.8	9.7	9.7 ^g	9.8 ^g	9.9	6.2	6.4	220	89	—	—	
8	Chicken minced	240	0	(42)	(0)	(39)	(100)	9.7	9.3 ^g	9.3 ^g	9.3	9.2	5.9	7.7	240	164	72	98	
9	Beef minced, 80% red meat	220	0	40	t	(51)	100	9.3	9.4	9.5 ^g	9.7 ^g	9.5	6.1	7.1	210	136	80	98	+
10	Pork minced	220	0	(40)	(0)	(44)	(100)	10.4	10.3 ^g	9.5 ^g	9.4 ^g	9.2	6.6	7.0	223	135	83	99	12.4
11	Fresh small sardine, m + s	400	0	37	t	14	100	6.6	9.2	9.4 ^g	9.4 ^g	9.5	6.2	6.6	220	248	141	97	+
12	Fresh common squid, m + s	240	0	(37)	(t)	(2)	(99)	9.9	10.0	9.8 ^g	9.8 ^g	9.4	6.6	6.6	218	194	43	99	
13	Short-necked clam	426	0	(35)	(5)	(4)	(87)	6.8	7.0 ^g	8.8 ^g	—	—	6.8	—	160	—	—	—	+
14	Skin of chicken	240	0	(32)	(0)	(101)	(100)	9.9	10.3 ^g	10.2 ^g	9.3	9.2	5.9	6.8	216	94	147	100	
15	Cod-surimi frozen	90	45	17	5	t	76	9.7	9.7 ^g	9.0 ^g	9.4	9.2	6.9	6.5	230	118	20	102	0.4
Plant-origin food																			
16	Wheat gluten	50	200					10.4	10.2	9.2	9.7	9.6	6.5	5.2	238	194	47	96	+
17	Sesame seeds, black	150	150	(30)	(28)	(81)	(52)	4.9	5.5 ^g	6.5 ^g	8.6	8.8	6.1	6.7	223	157	142	100	
18	Cocoa powder	120	120	29	65	12	31	3.6	3.5	3.9 ^g	4.5 ^g	4.1	5.3	7.2	215	131	106	99	
19	Soy sauce 230 and sugar 70	300	0	[17]	[86]	[0]	(17)	4.8	4.2 ^g	3.0 ^g	2.2	—	4.8	4.1	195	184	101	95	
20	Peppermint, dried leaf	100	200	16	52	3	47	5.8 ^g	3.4 ^g	3.8	4.8	—	6.2	8.4	184	216	71	96	

21	Coffee beans roasted	100	260	(13)	(56)	(16)	(18)	4.1 ^g	3.4	4.1 ^g	4.2 ^g	4.3	5.8	6.2	220	272	74	96
22	Sweet corn fresh	300	0	(10)	(60)	(4)	(14)	6.2	6.4	4.5 ^g	2.7 ^g	2.3 ^g	6.6	5.4	221	243	55	99
23	Sweet corn frozen	240	0	(8)	(48)	(3)	(14)	6.6	4.5 ^g	3.6 ^g	2.9 ^g	2.2	7.9	4.4	230	192	46	—
24	Garlic	80	80	(7)	(24)	(1)	(22)	6.5	6.1 ^g	3.9 ^g	3.1	—	7.1	5.1	180	113	41	96
25	Popcorn	62	150	(6)	(37)	—	(15)	4.0	3.9 ^g	3.4 ^g	2.9	2.9	5.9	5.6	210	154	35	89
26	Onion	300	0	3	15	t	15	5.3	3.5 ^g	2.3 ^g	2.2 ^g	2.6	5.2	6.9	238	269	18	—
27	Cabbage fresh	200	100	(3)	(11)	(1)	(20)	6.4	4.7	3.0 ^g	2.5 ^g	2.4	6.4	6.4	224	284	12	99
28	Tomato fresh	300	0	(2)	(11)	(1)	(16)	4.3 ^g	3.1 ^g	2.2 ^g	1.9	2.0	4.2	7.9	223	280	12	97
29	Sweet potato raw	200	50	(2)	(59)	(1)	(4)	3.4	2.5 ^g	2.2 ^g	2.2 ^g	2.6	6.2	5.8	218	175	50	90
30	Banana ripened	200	50	(2)	(46)	(1)	(5)	6.7	3.8	2.3 ^g	1.9 ^g	2.0	5.1	7.0	222	215	30	98
31	Fresh green alga ^h	240	0	—	—	—	—	4.6	3.2 ^g	2.4	2.6	4.0	7.0	7.2	235	199	25	93
32	Brown alga ⁱ frozen	240	0	—	—	—	—	4.1 ^g	4.5	6.4 ^g	5.4 ^g	4.3	6.2	7.4	226	191	42	97

Food mixed with both origins

33	Beef unheated, seasoned with vegetables and condiments, a manufactured product	220	0	34	16	18	68	6.8	4.9	3.7 ^g	3.7	4.3	5.6	5.5	230	138	77	98
34	Squid 200 seasoned with sugar 7, soy sauce 11, and a condiment "mirin" 24 (242 in total)	242	0	[32]	[17]	[2]	(64)	8.7	6.7 ^g	3.8 ^g	3.8	—	6.7	4.6	200	200	30	95
35	Simple "sukiyaki": beef 100, spring onion 100, soy sauce 10, sugar 15, sake 10, and a condiment "mirin" 5 (240 in total)	240	0	[21]	[95]	[47]	(46)	6.9 ^g	3.7 ^g	2.5 ^g	2.5 ^g	3.3	5.3	5.1	218	183	65	103
36	Minestrone canned, manufactured	240	0	5	11	2	31	6.5	4.8	3.3 ^g	3.1 ^g	—	5.1	5.4	205	217	16	97
37	Dried short-necked clam, manufactured: shucked by steam and seasoned with its drips and soy sauce	120	120	—	—	—	—	7.6	6.0	5.1	4.2	4.0	6.5	6.2	220	142	—	—

Other origin food

38	Mushroom frozen	300	0	(12)	(8)	(2)	(61)	10.2	9.4	9.0	8.8	8.7	7.3	7.4	230	266	18	95
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Note: Food of which fractionated distillates were, respectively, analyzed by GC-MS: 1 to 5, 9 to 11, 13, 15, 16, 23, 26, 33, and 37.

- ^a Content values without parentheses were obtained by determination. The values with parentheses were taken from the 4th ed. of the *Official Data of Food Components Analyzed* published in Japan.¹²⁷ The values with brackets were estimated by calculation on the basis of those data.
- ^b Water was added to some kinds of dried and juiceless food to help their volatiles distill much easily.
- ^c pH values were duplicated. The values of food before heating were directly measured or in the mixture with water. The ratio of sample to water was in the range of 1:1–1:9 (the data are available).
- ^d pH of food after heating was measured in the mixture of the thermal residue and glass beads with water in a ratio of 1:1 in almost all cases.
- ^e Handmade.
- ^f Trace indicates the content is less than 0.5%.
- ^g Fractions giving off characteristic pleasant heated-food odors.
- ^h *Ulva pertusa*.
- ⁱ *Dictyopteris prolifera*.

From Kawai and Komi⁴⁴ and Komi et al.¹⁰² With permission.

TABLE 13
Changes in Odors of Distillates from Animal- and Plant-Origin
Foods During Heating up to 230°C

Temperature range (°C)	Odor profiles	
	Animal-origin food	Plant-origin food
–99	Raw odor with boiled note	Raw odor with boiled odor
99–101	Boiled odor	Boiled odor
101–115	Well-boiled odor with roasted note	Well boiled with roasted note
115–160	Roasted odor with character- istic note	Roasted odor with ordinary Maillard reaction note
160–200	Well-roasted odor with some- what irritant odor	Well-roasted odor with burned sugar, cereal, or hay note
200–230	Charred odor with (NH ₄)CO ₃ ^a odor with irritant note	Charred odor with acidic note

^a Its powder often appears in the distillates nearly 200°C. It was isolated from chicken, pork, and dried squid.

From Kawai and Komi⁴⁴ and Komi et al.¹⁰² With permission.

proteinous wheat gluten (16) and sesame seeds (17), yielded acidic distillates in a range of pH 2 to 5. Foods consisting of both origins gave relatively lower pHs of the distillates at the beginning of heating than those of animal-origin foods, as exemplified by two comparisons of sukiyaki (35) to beef (9), and seasoned squid (34) to fresh squid (12).

From a comparison of pH in the distillate to the major component contents, the basic and acidic pHs were largely dependent on the high protein and carbohydrate contents, respectively. An exceptional pH alteration of sesame seeds (17), which showed a gradual rise with an increase in temperature, was probably due to nearly the same contents of protein and carbohydrate.

D. Thermal Environments of Foods

The first fraction obtained from nearly 100°C is almost the same as a distillate produced by an ordinary steam distillation method for GC analysis. Many GC data must have been influenced by pH as shown in Table 12.

Thermal environments are classified into two types: basic and acidic. The former belongs to proteinous foods and the latter to protein-less foods. The basic environment is due to the gen-

eration of basic low-bp compounds. The low-bp compound that elevates the pH of distillates is probably ammonia, as described in Section II.E. In addition to the generation of ammonia from seafoods, the nonenzymatic deamidation of amide residues in soy and wheat flour proteins was reported recently: more ammonia was produced at higher temperatures and at a higher pH.^{103,104} The amount of ammonia liberated from high-gluten wheat flour (14% protein) at pH 13.0 and 40°C was reported as that reaching nearly 540 mg/100 g flour.¹⁰⁴ Thus, high pHs of distillates from the tested proteinous foods in Table 12 probably result from ammonia released from the proteins degraded during heating. Exceptionally, the delay of increase in pH was observed in the short-necked clam (13):⁴¹ there was no significant difference in the pH values of the clam juice (pH 6.8) between the first distillate (pH 6.8) and the second one (pH 7.0). It supports the fact that neither fresh nor boiled short-necked clam contained ammonia.¹⁰⁵

Generally, odors arising from heated animal-origin foods are more unstable during storage than those arising from plant-origin foods. The instability of roasted food odors obtained from the distillates seems partly responsible for the degradation of aldehydes under the influence of high pH (see Figure 4). Low fatty aldehydes are common compounds in proteinous food volatiles.⁴⁶

They have often been regarded as Strecker degradation products of amino acids.¹⁰⁶ However, the Strecker degradation products need a higher temperature (100°C) and the presence of sugar for production of aldehydes,¹⁰⁷ than retro-aldol condensation like that under pH 11 at 21°C (Figure 4). The retro-aldol condensation was also more accelerated by higher temperatures.⁶³ Small amounts of carbohydrates in the proteinous foods (in Table 12) seem doubtful whether Strecker reactions occur with little heating of proteinous foods at near 100°C. Low fatty aldehydes detected from the proteinous foods are likely derived from the lipids through retro-aldol condensation.

As Table 12 shows the thermal conditions for sardine (2), the volatiles of sardine in Table 9 must have been formed under the basic (pH 9.0) and high-temperature conditions. Compared with the volatiles of sardine shown in Figure 5 and in Table 9, the thermal basic conditions allowed 4Z-heptenal to be produced in the first distillate of sardine. Also, the high pH (10.3) and the tempera-

ture (100°C) observed in cod muscle with skin (4) might have allowed the aldehyde to be formed in the steam distillate of cod fillet.⁷⁶ Unheated extraction (21°C) of salmon B in Table 3 produced 4Z-heptenal along with other unsaturated aldehydes, suggesting that salmon B and the analogs in Table 4 are specific.

There was no difference in pH alteration of the distillates between the groups of seafood and other meats. The former must have been influenced at least by ammonia and TMA, and the latter ammonia. The pH cannot distinguish salt- and freshwater fishes. The acidic conditions of plant volatiles were derived from the fatty acids produced from the Maillard reaction of sugars with amino acids.

As described above, pH affects odor quality and generation of volatiles: mixtures of TMA and acids (Table 1), whitefish extracts (Table 10), *S,N*-heterocycles (Table 14), retro-aldol formation of 4Z-heptenal (Figure 4), and H₂S formation (Figure 6). Under the influence of pH, volatile constituents are formed.

TABLE 14
Organoleptic Properties of 3,4,5,6-tetrahydro-2,4,6-trialkyl-2H-1,3,5-thiadiazines (TD) and 5,6-Dihydro-2,4,6-trialkyl-4H-1,3,5-dithiazines (DT) at Different pH Values and Equivalent Points (*Ep*)

Compound	<i>Ep</i>	pH	Odor profiles
Methyl-TD	4.0	9.1 ^a	Strong roast cereal
		5.5	Strong roast popcorn
		4.0	Medium ammonium sulfide
		1.4	Strong ammonium sulfide
Ethyl-TD	6.4	9.0 ^a	Weak avocado, raw pumpkin, unripened orange
		6.1	Odor added with that of green vegetable
		4.9	Strong fresh leek, slightly earthy
		3.0	Very strong crushed fresh leek
		1.9	Weak staled mustard
Methyl-DT	2.5 ^b	8.1 ^a	Medium roast shrimp
		3.5	Medium roast shrimp
		2.7	Medium roast shrimp with slight sweetness
		1.6	Weak edible mushroom "shiitake"
		7.3 ^a	Weak raw pumpkin, white parts of Allium plants
Ethyl-DT	1.7	5.5	Odor added with green note
		4.5	Strong leek
		2.1	Medium leek, onion
		0.4	Weak Allium plants

^a Before addition of dilute HCl.

^b 2.7¹²⁶.

From Kawai et al.⁹⁴ With permission.

E. Thermal Inner Situations of Foods

The large gap of pHs between a distillate and its own food in Table 12 indicates that there is no relationship between them, suggesting that the situations between an atmosphere surrounding food and an inner part of food are independent of each other. The former was due to volatiles, and the latter to some nonvolatile pH-buffer substances. Thus, pH separates thermal food environments into volatile and nonvolatile parts, outer and inner situations, that is, food circumstances are influenced by pH (or volatiles) and the inner situations are independent of pH. The concept may explain why microwave heating does not much produce roasted-food odors: the microwave method heats the inner parts of foods, where stable pH prevents low-molecular-weight volatiles from forming roasted-food odors. The shortcomings of microwave-cooked foods are that they do not brown in the process and flavors are not enhanced,⁴⁰ because of the stable pH of the inner parts of foods.

There is a fundamental difference between aroma and taste. Taste is more stable than aroma

under thermal conditions. For example, the non-volatile components of crab leg meat, namely, ionic compounds (Cl^- , Na^+ , K^+ , and PO_4^{3-}), amino acids, nucleotides, nucleosides, betaine, and sugars, showed almost unchanged variations during heating the meat up to 100°C except for a temperature range of 75 to 85°C ,¹⁰⁸ where the coagulation of crab protein may have occurred. Recently, efficient taste-active components in seafoods for preparing artificial taste combined by nonvolatile pure chemicals were reviewed, as represented in Table 15.¹⁰⁹ Some ionic components in the list, together with glutamic acid and glycine, are well recognized as common taste activators, whereas the ionic component tastes were strongly dependent on pH. Even the synthetic tastes were influenced by pH, irrespective of their nonvolatility.¹⁰⁹ The factor stabilizing the pH of heated foods is unknown, but possibly it was due to large molecular weight compounds such as proteins.

TABLE 15
Taste-Active Components in Seafoods (mg/100 g)

	Abalone	Sea urchin	Snow crab	Scallop	Short-necked clam
Glutamic acid	109^a	103	19	140	90
Glycine	174	842	623	1925	180
Alanine	98	261	187	256	74
Valine	37	154	30	8	4
Methionine	13	47	19	3	3
Arginine	299	316	579	323	53
Taurine	946	105	243	784	555
AMP	90	10	32	172	28
IMP	— ^b	2	5	—	—
GMP	—	2	4	—	—
Betaine	975	7	357	339	42
Succinic acid	—	1.2	9	10	65
Na^+	— ^c	—	191	73	244
K^+	—	—	197	218	273
Cl^-	—	—	336	95	322
PO_4^{3-}	—	—	217	213	74

Note: AMP, adenosine 5'-monophosphate; IMP, inosine 5'-monophosphate; GMP, guanosine 5'-monophosphate.

^a Taste-active components are shown in bold numbers.

^b Not detected.

^c Not analyzed.

From Fuke and Konosu.¹⁰⁹ With permission.

F. Effect of pH on Formation of *S,N*-Heterocycles

Table 12 also exhibits the *S,N*-heterocycles (DT, TD, and PD) found in the distillates submitted to GC analysis. All foods that produced at least one of the *S,N*-heterocycles are proteinous (Table 12). On the other hand, every plant- and both-origin food gave no *S,N*-heterocycles except for proteinous wheat gluten, indicating that the compounds were formed under basic conditions. Due to the characteristic odor properties, the *S,N*-heterocycles are contributors to roasted-food odors.^{43,95,96} The compounds are nonenzymatically produced during heating, resulting from interaction of ammonia, aldehydes, and H₂S. To reveal the significance of thermal environments of food, we review here some proposed formation mechanisms for the *S,N*-heterocycles.

As a representative of the chemical class of TDs, 5,6-dihydro-2,4,6-trimethyl-4*H*-1,3,5-dithiazine (thialdin) was first isolated from a mixture of hexahydro-2,4,6-trimethyl-1,3,5-triazine three hydrates (methyl TZ, a trimer of 1-amino-1-ethanol monohydrate) and H₂S by Wöhler and von Liebig in 1814,¹¹⁰ after von Liebig isolated the TZ from a mixture of ethanal and ammonia¹¹¹ (see Figure 9). This methyl DT, thialdin, was registered in Beilsteins Handbook as system no. 4397. Later, it was discovered as a major natural volatile in heated beef broth.¹¹² After that, as

the major volatile, it has been detected in many proteinous foods: antarctic krill,⁹⁵ small shrimp,^{19,113} dried squid,⁴³ dried short-necked clam,¹¹⁴ dry red bean,¹¹⁵ and yeast extract.¹¹⁶

Methyl DT and TD were obtained from a mixture of ammonia and *bis*(1-mercaptoethyl)sulfide prepared from ethanal and H₂S by Boelens et al.¹¹⁷ They proposed a mechanism for forming these compounds as shown in Figure 10, where the reaction of ethanal with H₂S precedes that of ethanal with ammonia. The compounds in Figure 10 and the conditions for their formation were considered extremely important for the production of desirable meat flavors.¹¹⁸ After that, and simultaneously with Kawai et al.,⁹⁴ Shu et al.¹¹⁹ proposed another mechanism of formation for DT (see Figure 11). This mechanism was substantially similar to the one above: (1) an aldehyde reacts with H₂S first and then ammonia, and (2) a 3,5-dialkyl-1,3,5-trithiolane and DT arise from similar intermediates almost simultaneously. Werkhoff et al.⁹⁶ also reported that the reaction of aldehydes with H₂S preceded that of ammonia (see Figure 12).

On the other hand, Kawai et al.⁹⁴ proposed a different formation mechanism (see Figure 13) in which the compounds were proved by the synthetic procedures in each step from aldehydes to DTs. The reversible relationship between the compounds was also confirmed. As Figure 13 shows, DTs are formed from TZs through TDs by two

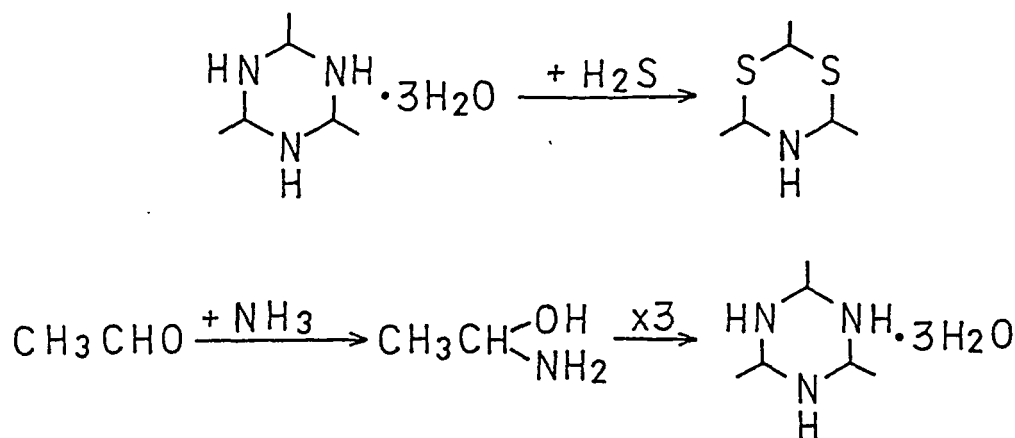


FIGURE 9. Synthetic formation of 5,6-dihydro-2,4,6-trimethyl-4*H*-1,3,5-dithiazine from hexahydro-2,4,6-trimethyl-triazine and H₂S, and that of the triazine from ethanal and ammonia. (From Wöhler and von Liebig.¹¹⁰ With permission.)

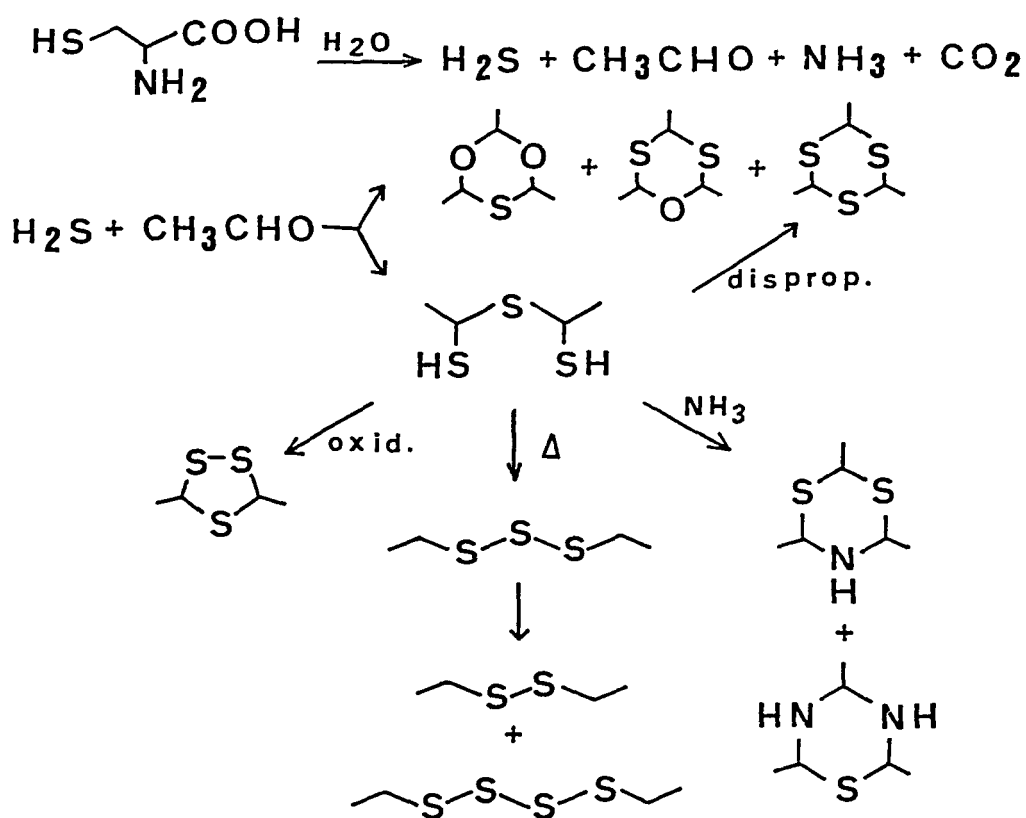


FIGURE 10. Possible mechanism for the formation of 5,6-dihydro-2,4,6-trimethyl-4H-1,3,5-dithiazine, 3,4,5,6-tetrahydro-2,4,6-trimethyl-2H-1,3,5-thiadiazine, 3,5-dimethyl-1,2,4-trithiolane, and other compounds through bis(1-mercaptoethyl)sulfide. (From Boelens et al.¹¹⁷ With permission.)

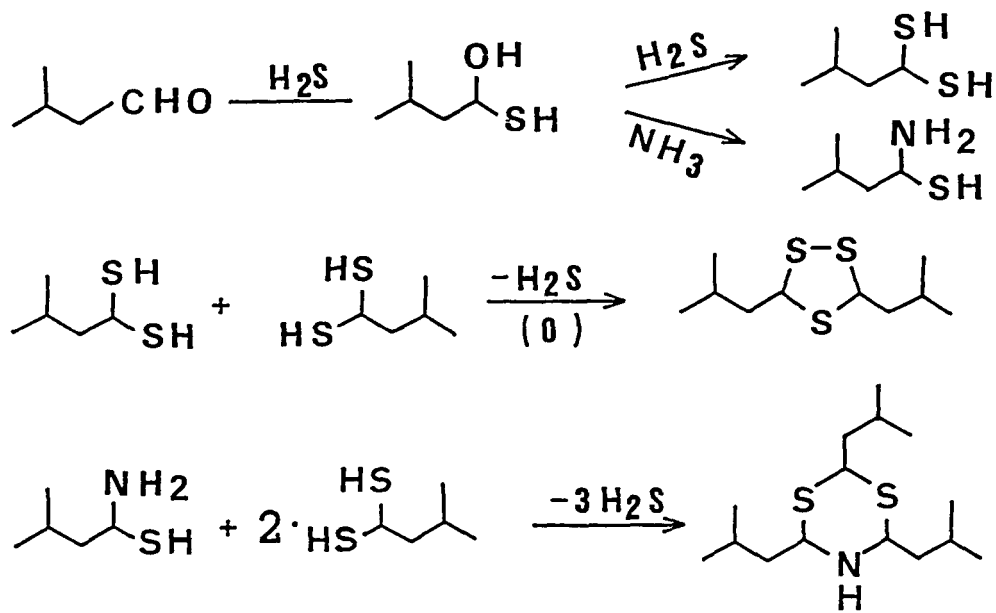


FIGURE 11. Possible mechanism of the formation of 3,5-diisobutyl-1,2,4-trithiolane and 5,6-dihydro-2,4,6-triisobutyl-4H-1,3,5-dithiazine. (From Shu et al.¹¹⁹ With permission.)

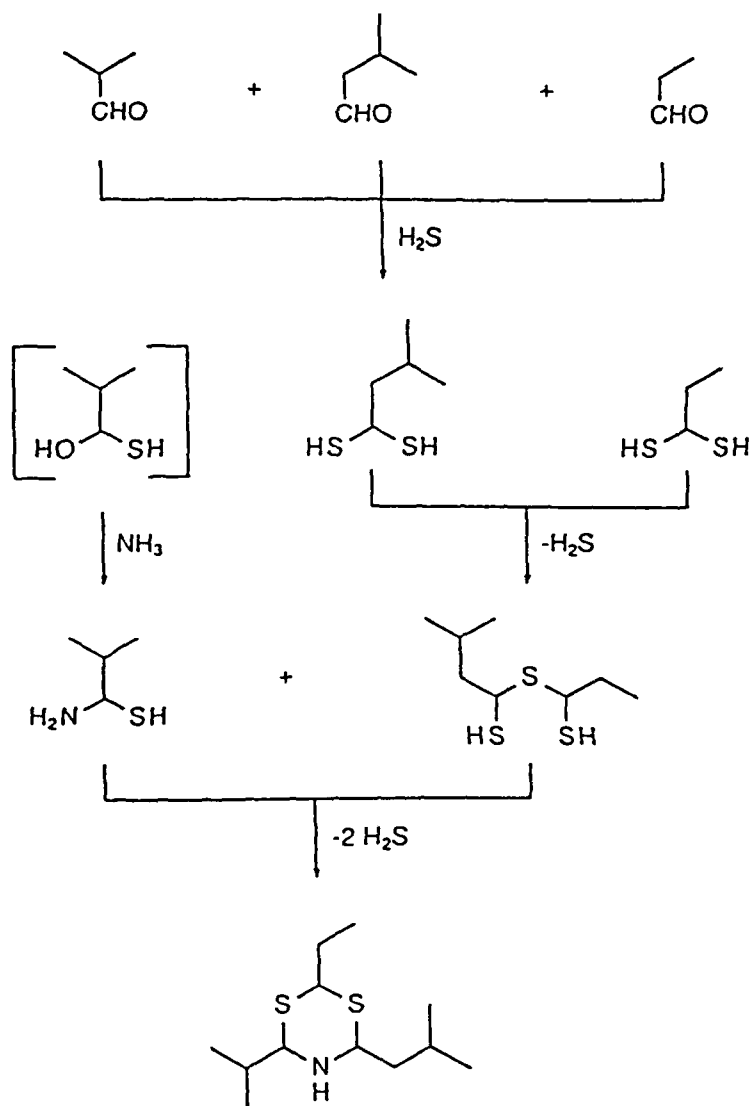
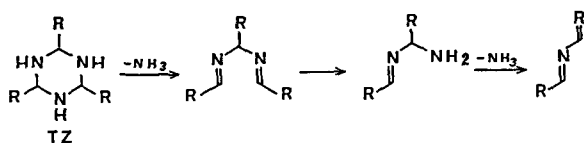


FIGURE 12. Possible mechanism for the formation of 2-ethyl-4-isobutyl-6-isopropyl-5,6-dihydro-4H-1,3,5-dithiazine. (From Werkhoff et al.⁹⁶ With permission.)

steps of addition of H_2S . If H_2S is not enough at this stage, TDs are spontaneously degraded into DTs and N,N' -dialkylidene-1,1-diaminoalkanes, which are further degraded into lower N-compounds, N -alkylidene-1,1'-diaminoalkanes, and finally to N -alkylidene-1-amino-1-alkenes.



SCHEME 4. Degradation of TZ.

It is remarkable that the mechanism of formation occurs under basic conditions: ammonia first reacts with ethanal, after which H_2S stepwise superimposes on the resultant methyl TZ to produce methyl TD and further methyl DT in high yields; methyl TD dihydrates 70% and methyl DT 95% yields^{41,43} without producing 3,5-dialkyl-1,2,4-trithiolanes.⁴² In contrast with methyl DT, the trithiolane formed largely under acidic pH ranges of thermal aqueous S-containing amino acids releasing ethanal, ammonia, and H_2S , for example, cystine at pH 2.3¹²⁰ and cysteine at pH 5.1.¹²¹

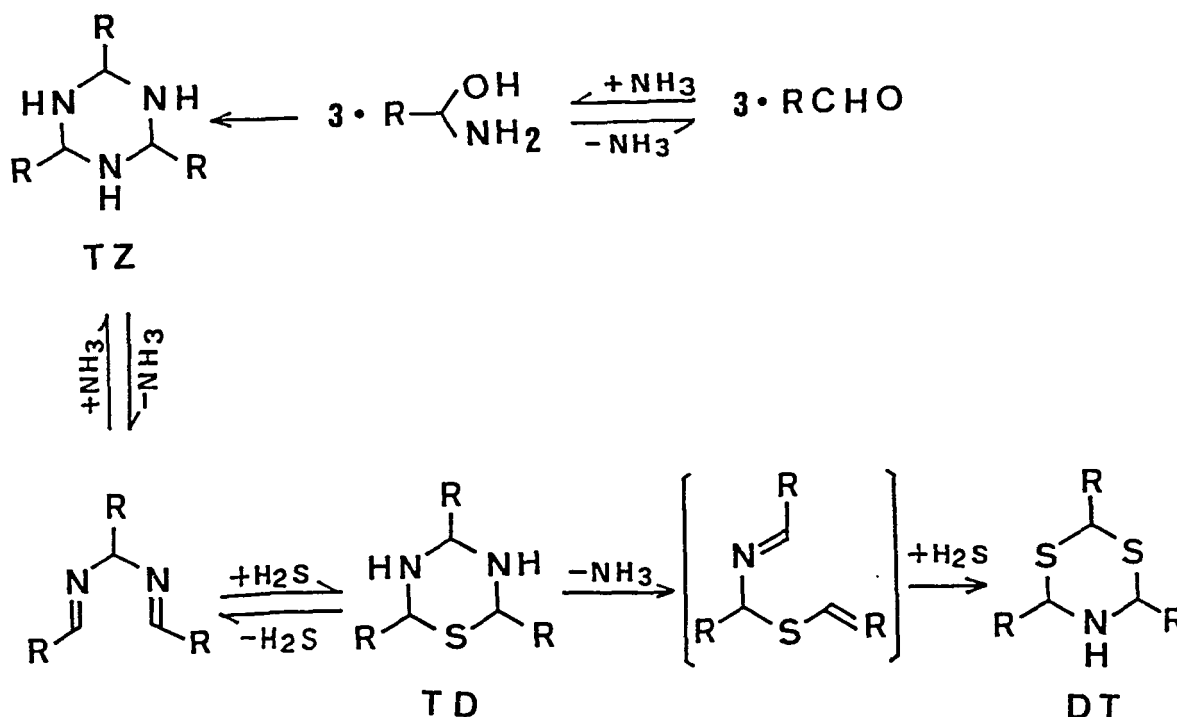


FIGURE 13. Possible mechanism for the formation of 5,6-dihydro-2,4,6-trialkyl-4H-1,3,5-dithiazines (DT) through 3,4,5,6-tetrahydro-2,4,6-trialkyl-2H-1,3,5-thiadiazines (TD). (From Kawai et al.⁹⁴ With permission.)

Under neutral conditions (pH 7.5), cysteine yielded more methyl DT than 3,5-dimethyl-1,2,4-trithiolane.¹²² Further, the GC peak of the trithiolane became vague with an increase in pH.¹²³ Those results indicated that two kinds of formation of methyl DT and the trithiolane do not occur simultaneously except in neutral pH ranges, suggesting that formation mechanisms of methyl DT and the trithiolane are independent of each other.

As seen in Figure 10, *bis*(1-mercaptoethyl)sulfide was proposed as a mutually common precursor of methyl DT and the trithiolane. However, it is doubtful that the unstable sulfide to an alkali occurs largely from ethanal and H₂S against the basic conditions as exhibited in the distillates of proteinous foods in Table 12, and also the sulfide becomes superior to N-compounds formed from ethanal and ammonia. For the same reason, possible mechanisms of DTs in Figures 11 and 12 are also doubtful for natural DTs. Many DTs identified as food volatiles are considered to have been formed through the pathways in Figure 13. According to the mechanism, elimination of ammonia and addition of H₂S occur stepwise in

the mechanism where some *S,N*-heterocycles reported as degradation products of cysteine¹²² were found.

The thermal environment of dried squid (Figure 7 and Table 11) was reported to have produced N-compounds (28% of the GC area in total), *S,N*-heterocycles (9%), and S-compounds (2%) under basic conditions.⁴³ The amount of *S,N*-heterocycles produced was in almost agreement with that shown in Figure 13 as follows: (1) ammonia largely occurs prior to H₂S (Figure 7) in the same manner in fresh mackerel as described in Section II.E,⁹¹ (2) under the basic conditions by large amounts of ammonia and TMA,⁴² ordinary aldehydes are rapidly formed through retro-aldol condensation (or autoxidation) at relatively low temperature (Figure 4); (3) ammonia reacts with aldehydes to form TZs (Figure 13); finally, (4) TZs react with H₂S of which formation is accelerated by basic pH (Figure 6), and DTs are formed through TDs as final reaction products.⁹⁴

Almost all the foods reported as having the *S,N*-heterocycles in the volatile constitution must have belonged to proteinous foods that produce

the basic distillates in the same way as those in Table 12. However, as Table 12 shows, in spite of basic distillates, there were differences in the detection of the *S,N*-heterocycles among the distillates loaded on GC-MS. The detection was limited in the fractionated distillates that showed a tendency to decrease in pH during heating, such as dried squid (3), dried short-necked clam (5), and pork (10). The tendency is consistent with the formation mechanism in Figure 13. There was no detection of *S,N*-heterocycles despite highly basic distillates from fresh cod muscle (4), suggesting that there was too much ammonia to form DTs, and instead produced many N-compounds, such as indole, pyrrole, and *N,N*-dimethylaminonitrile. Further, each fraction of heated cod muscle gave a different volatile constitution from cod fillet⁷⁶ and cod liver oil.⁵⁷ Thus, fresh cod muscle, together with sardine (2), emits almost no unsaturated carbonyls and alcohols even after heating, suggesting that no enzymatic reaction occurs while cod is fresh enough.

Concerning formation of DTs in domestic animal meats and chicken muscle, we reported previously that neither high pH nor a remarkable increase in pH during boiling^{29,124} could cause DT, especially DT substituted with higher alkyl groups, to arise from such food volatiles.⁴² However, as Table 12 shows, all proteinous foods, including beef, pork, and chicken, gave basic distillates even from initial stage of heating, independent of pH in thermal residues. Therefore, we stand corrected.

IV. CONCLUSION

Recent studies on enzymatic and nonenzymatic oxidation of PAs concerning euryhaline fish flavor have revealed characteristic volatiles and the mechanisms of formation: unsaturated carbonyl and alcohol compounds derived from oxidation of PAs have been found in many kinds of euryhaline fish, especially anadromous fish such as salmon and trout at the prime season. Its variety of volatiles consisting mainly of C₈- and C₉-unsaturated compounds belongs to euryhaline fish characteristically. The variety is probably enhanced by spawning and osmoregulation conditions as described for the mature salmon migrat-

ing from the sea. It is in contrast to nearly monotonous volatile constitution of sardine, cod, and other saltwater fishes.

For saltwater fish, the fact that any oxidation of PAs does not occur is possibly more responsible for an unknown antioxidation system than difference in enzymatic reactivities toward PAs, because saltwater fish have an enzymatic system similar to that of euryhaline fish. Generation of ammonia in large quantity prior to that of any other compounds should be significant to produce its succeeding flavor compounds, which depend on freshness. Ammonia contributes little to the odor due to its high organoleptic threshold value. Large amounts of ammonia, however, play an important role in changing the basic environment, which accelerates generation of other basic compounds like TMA. Ammonia reacts with aldehydes easily, producing more complex flavor components such as *S,N*-heterocycles during the heating process, and subsequently dominating over volatile constitution and odor quality. TMA, regarded as the most characteristic flavor compound for saltwater fish, also affects the odor quality under the basic environment, as exemplified in combination with fatty acids. Further research on combinations of TMA with carbonyls and alcohols is needed to confirm saltwater fish flavor.

For freshwater fish, its characteristic constituents are likely responsible for pyrrolidine and piperidine. Some model reactions of these compounds or their precursors, 1,4-diaminobutane and 1,5-diaminopentane, with carbonyls or alcohols are expected to be performed for accumulation of the MS data. Possibly due to the lack of data, a small number of identifiable compounds might have remained.

Thermal environments of seafood, which are indicated by the pH of distillates arising from heated foods, can be classified into basic and acidic, in response to the high and low protein contents, respectively. The pH in distillates is independent of the pH in food, suggesting the different concepts between an odor and a taste. Because all the tested seafoods produce basic distillates in the same way as domestic meats and proteinous plant foods, the pH cannot distinguish seafood from others. pH alteration, however, allows estimation of the formation of *S,N*-heterocycles as flavor components in the distillates. The

heterocycle formation indicates the generation of ammonia, aldehydes, and H₂S, and their interactions.

REFERENCES

1. Maarse, H. and Visscher, C. A., Eds., *Volatile Compounds in Food—Qualitative and Quantitative Data. Supplement 2 and Cumulative Index*, TNO Biotechnology and Chemical Institute, Zeist, The Netherlands, 1991.
2. Ikeda, S., Other organic components and inorganic components, in *Advances in Fish Science and Technology*, Connell, J. J., Ed., Fishing News Books, Surrey, England, 1980, 111.
3. Yoshinaka, R., Volatile components (of fish and shellfish), in *Micro Components of Fish and Shellfish (in Japanese)*, Ikeda, S., Ed., Koseisha-Koseikaku, Tokyo, 1981, 110.
4. Tokunaga, T., Chemistry of fish odor, in *Fish and Meat Odor (in Japanese)*, Ohta, S., Ed., Koseisha-Koseikaku, Tokyo, 1981, 29.
5. Reineccius, G., Off-flavors in foods, *Crit. Rev. Food Sci. Nutr.*, 29, 381, 1991.
6. German, J. B. and Kinsella, J. E., Lipid oxidation in fish tissue. Enzymatic initiation via lipoxygenase, *J. Agric. Food Chem.*, 33, 680, 1985.
7. Hsieh, R. J., German, J. B., and Kinsella, J. E., Lipoxygenase in fish tissue: some properties of the 12-lipoxygenase from trout gill, *J. Agric. Food Chem.*, 36, 680, 1988.
8. Josephson, D. B. and Lindsay, R. C., Enzymic generation of volatile aroma compounds from fresh fish, in *Biogenesis of Aroma*, Parliment, T. H. and Croteau, R., Eds., ACS Symposium Series 317, American Chemical Society, Washington, D.C., 1986, 201.
9. Karahadian, C. and Lindsay, R. C., Role of oxidative processes in the formation and stability of fish flavor, in *Flavor Chemistry: Trends and Developments*, Teranishi, R., Buttery, R. G., and Shahidi, F., Eds., ACS Symposium Series 388, American Chemical Society, Washington, D.C., 1989, 60.
10. Josephson, D. B., Lindsay, R. C., and Stuibler, D. A., Volatile compounds characterizing the aroma of fresh Atlantic and Pacific oysters, *J. Food Sci.*, 50, 5, 1985.
11. Hirano, T., Fish, in *Odor of Marine Products*, Koizumi, C., Ed., Koseisha-Koseikaku, Tokyo, 1989, 31.
12. Lindsay, R. C., Fish flavors, *Food Rev. Int.*, 6, 437, 1990.
13. Josephson, D. B., Seafood, *Food Sci. Technol.*, 41, 179, 1991.
14. Sakakibara, H. and Hayashi, K., Analysis of fish odors, *Yukagaku*, 38, 848, 1989.
15. Sakakibara, H., Hosokawa, M., Yajima, I., and Hayashi, K., Flavor constituents of dried bonito (Katsuobushi), *Food Rev. Int.*, 6, 553, 1990.
16. Whitfield, F. B., Flavor of prawns and lobsters, *Food Rev. Int.*, 6, 505, 1990.
17. Iida, H., Shellfish, in *Odor of Marine Products*, Koizumi, C., Ed., Koseisha-Koseikaku, Tokyo, 1989, 42.
18. Kasahara, K., Cooked and processed fish, in *Odor of Marine Products*, Koizumi, C., Ed., Koseisha-Koseikaku, Tokyo, 1989, 93.
19. Kubota, K., Uchida, T., Kurosawa, K., Komura, A., and Kobayashi, A., Identification and formation of characteristic volatile compounds from cooked shrimp, in *Thermal Generation of Aromas*, Parliment, T. H., McGorin, R. J., and Ho, C.-T., Eds., ACS Symposium Series 409, American Chemical Society, Washington, D.C., 1989, 376.
20. Hsieh, T. C.-Y., Vejaphan, W., Williams, S. S., and Matiella, J. E., Volatile flavor compounds in thermally produced Louisiana red swamp crayfish and blue crab, in *Thermal Generation of Aromas*, Parliment, T. H., McGorin, R. J., and Ho, C.-T., Eds., ACS Symposium Series 409, American Chemical Society, Washington, D.C., 1989, 386.
21. Noda, H., Seaweed products, in *Odor of Marine Products*, Koizumi, C., Ed., Koseisha-Koseikaku, Tokyo, 1989, 83.
22. Sugisawa, H., Nakamura, K., and Tamura, H., The aroma profile of the volatiles in marine green algae, *Food Rev. Int.*, 6, 573, 1990.
23. Kajiwarra, T., Kodama, K., Hatanaka, A., and Matsui, K., Volatile compounds from Japanese marine brown algae, in *Bioactive Volatile Compounds from Plants*, Teranishi, R., Buttery, R. G., and Sugisawa, H., Eds., ACS Symposium Series 525, American Chemical Society, Washington, D.C., 1993, 103.
24. Reineccius, G. A., Symposium on meat flavor. Off-flavor in meat and fish. A review, *J. Food Sci.*, 44, 12, 1979.
25. Yamanaka, H., Offensive odor of fish and shellfish, in *Odor of Marine Products*, Koizumi, C., Ed., Koseisha-Koseikaku, Tokyo, 1989, 53.
26. Schultz, H. W., Day, E. A., and Libbey, L. M., Eds., *Symposium on Foods: The Chemistry and Physiology of Flavors*, AVI Publishing Company, Westport, CT, 1967.
27. Jones, N. R., Fish flavor, in *Symposium on Foods: The Chemistry and Physiology of Flavors*, Schultz, H. W., Day, E. A., and Libbey, L. M., Eds., AVI Publishing Company, Westport, CT, 1967, 267.
28. Kikuchi, T., Wada, S., and Suzuki, H., Significance of volatile bases and volatile acids in the development of off-flavor of fish meat, *Eiyo to Shokuryo*, 29, 147, 1976.
29. Johnson, A. R. and Vickery, J. R., Factors influencing the production of hydrogen sulfide from meat during heating, *J. Sci. Food Agric.*, 15, 695, 1964.
30. Hamm, R. and Hofmann, K., Changes in the sulphydryl and disulfide groups in beef muscle proteins during heating, *Nature (London)*, 207, 1269, 1965.

31. Ishida, Y., Physiological studies on evolution of dimethyl sulfide from unicellular marine algae, *Mem. Coll. Agric. Kyoto Univ.*, 94, 47, 1968.
32. Iida, H., Studies on the accumulation of dimethyl- β -propiothetin and the formation of dimethyl sulfide in aquatic organisms, *Bull. Tokai Res. Fish. Res. Lab.*, 124, 35, 1988.
33. Tokunaga, T., On the thermal decomposition of trimethylamine oxide in muscle of some marine animals, *Bull. Jpn. Soc. Sci. Fish.*, 52, 557, 1986.
34. Namiki, M. and Hayashi, T., A new mechanism of the Maillard reaction involving sugar fragmentation and free radical formation, in *The Maillard Reaction in Foods and Nutrition*, Waller, G. R. and Feather, M. S., Eds., ACS Symposium Series 215, American Chemical Society, Washington, D.C., 1983, 21.
35. Einarsson, H., The effect of time, temperature, pH and reactants on the formation of antibacterial compounds in the Maillard reactions, *Lebensm.-Wiss. u.-Technol.*, 20, 51, 1987.
36. Güntert, M., Brüning, J., Emberger, R., Hopp, R., Köpsel, M., Surburg, H., and Werkhoff, P., Thermally degraded thiamine. A potent source of interesting flavor compounds, in *Flavor Precursors: Thermal and Enzymatic Conversion*, Teranishi, R., Takeoka, G. R., and Güntert, M., Eds., ACS Symposium Series 490, American Chemical Society, Washington, D.C., 1992, 140.
37. Shu, C. K. and Ho, C.-T., Effect of pH on the volatile formation from the reaction between cysteine and 2,5-dimethyl-4-hydroxy-3(2H)-furanone, *J. Agric. Food Chem.*, 36, 801, 1988.
38. Hayase, F. and Kato, H., Low-molecular Maillard reaction products and their formation mechanisms, *Dev. Food Sci.*, 13, 39, 1986.
39. de Roos, K. B., Meat flavor generation from cysteine and sugars, in *Flavor Precursors: Thermal and Enzymatic Conversions*, Teranishi, R., Takeoka, G. R., and Güntert, M., Eds., ACS Symposium Series 490, American Chemical Society, Washington, D.C., 1992, 203.
40. Yeo, H. C. H. and Shibamoto, T., Microwave-induced volatiles of the Maillard model systems under different pH conditions, *J. Agric. Food Chem.*, 39, 370, 1991.
41. Kawai, T., Ishida, Y., Kakiuchi, H., Ikeda, N., and Tsuneya, T., Volatile components of boiled and roasted short-necked clam *Tapes philippinarum*, *Bull. Jpn. Soc. Sci. Fish.*, 56, 795, 1990.
42. Kawai, T., Thiadiazines and Dithiazines in Volatiles from Heated Seafoods, and Mechanisms of Their Formation, Ph.D. dissertation, Kyoto University, 1990.
43. Kawai, T., Ishida, Y., Kakiuchi, H., Ikeda, N., Higashida, T., and Nakamura, S., Flavor components of dried squid, *J. Agric. Food Chem.*, 39, 770, 1991.
44. Kawai, T. and Komi, H., Changes in pH of distillates from heating food to 230°C, and the relationship to heated food volatiles, in *Abstracts of IFTEC International Food Technology Exposure and Conference*, The Hague, The Netherlands, 1992, 200.
45. Wong, N. P., Damico, J. N., and Salwin, H., Decomposition and filth in foods: investigation of volatile compounds in cod fish by gas chromatography and mass spectrometry, *J.A.O.A.C.*, 50, 8, 1967.
46. Maarse, H. and Visscher, C. A., Eds., *Volatile Compounds in Food, Qualitative Data, Supplement 5*, TNO-CIVO Food Analysis Institute, Zeist, The Netherlands, 1988.
47. Obata, Y., Yamanishi, T., and Ishida, M., Chemical studies on the substance of fish smell; piperidine group compounds as the substance concerned, *Bull. Jpn. Soc. Sci. Fish.*, 15, 551, 1950.
48. Obata, Y. and Yamanishi, T., Chemical studies on the substance of fish smell; investigation on the components of mucilaginous substances of salmon, *Bull. Jpn. Soc. Sci. Fish.*, 15, 554, 1950.
49. Shimakura, K., Yamanaka, H., Shiomi, K., and Kikuchi, T., Gas liquid chromatographic determination of pyrrolidine and piperidine in canned fish products, *Bull. Jpn. Soc. Sci. Fish.*, 57, 1753, 1991.
50. Yamanaka, H., Shimakura, K., Shiomi, K., Kikuchi, T., Iida, H., and Nakamura, K., Concentration of polyamines in freshwater fishes, *Bull. Jpn. Soc. Sci. Fish.*, 53, 2041, 1987.
51. Tsuchiya, Y. and Matsumoto, A., Identification of volatile metabolites produced by blue-green algae, *Water Sci. Technol.*, 20, 149, 1988.
52. Mills, O. E., Chung, S.-Y., and Johnson, P. B., Dehydration products of 2-methylisoborneol are not responsible for off-flavor in the channel catfish, *J. Agric. Food Chem.*, 41, 1690, 1993.
53. Yagi, M., Odorous compounds in raw water of water supplies (A review), *Jpn. J. Water Pollut. Res.*, 18, 704, 1985.
54. Josephson, D. B., Lindsay, R. C., and Stuijber, D. A., Influence of maturity on the volatile aroma compounds from fresh Pacific and Great Lakes salmon, *J. Food Sci.*, 56, 1576, 1991.
55. Josephson, D. B., Lindsay, R. C., and Stuijber, D. A., Variation in the occurrences of enzymically derived volatile aroma compounds in salt- and freshwater fish, *J. Agric. Food Chem.*, 32, 1344, 1984.
56. Crawford, L., Kretsch, M. J., and Duaoagni, D., Identification of volatiles from tuna fish, *J. Sci. Food Agric.*, 27, 531, 1976.
57. Karahadian, C. and Lindsay, R. C., Evaluation of compounds contributing characterizing fishy flavors in fish oils, *J. Am. Oil Chem. Soc.*, 66, 953, 1989.
58. Hsieh, T. C. Y., Williams, S. S., Vejaphan, W., and Meyers, S. P., Characterization of volatile components of menhaden fish (*Brevoortia tyrannus*) oil, *J. Am. Oil Chem. Soc.*, 66, 114, 1989.
59. Hatanaka, A., Kajiwar, T., and Sekiya, J., Fatty acid hydroperoxide lyase in plant tissues. Volatile aldehyde formation from linoleic and linolenic acids, in *Biogenesis of Aromas*, Parliment, T. and Croteau, R., Eds., ACS Symposium Series 317,

- American Chemical Society, Washington, D.C., 1986, 167.
60. Zhang, C.-H., Hirano, T., Suzuki, T., and Shirai, T., Enzymatically generated specific volatile compounds in ayu tissue, *Bull. Jpn. Soc. Sci. Fish.*, 58, 559, 1992.
 61. Mohri, S., Cho, S.-Y., Endo, Y., and Fujimoto, K., Linoleate 13(S)-lipoxygenase in sardine skin, *J. Agric. Food Chem.*, 40, 573, 1992.
 62. Cho, S.-Y., Mohri, S., Endo, Y., and Fujimoto, K., Characterization of lipid prooxidants in sardine skin, *Bull. Korean Fish Soc.*, 25, 501, 1992.
 63. Josephson, D. B. and Lindsay, R. C., Retro-aldol degradation of unsaturated aldehydes: role in the formation of c4-heptenal from *12,c6*-nonadienal in fish, oyster and other flavors, *J. Am. Oil Chem. Soc.*, 64, 132, 1987.
 64. Suyama, M., Hirano, T., and Yamazaki, S., Odor of ayu and its volatile components, *Bull. Jpn. Soc. Sci. Fish.*, 51, 287, 1985.
 65. Hirano, T., Zhang, C.-H., Morishita, A., Suzuki, A., and Shirai, T., Identification of volatile compounds in ayu fish and its feeds, *Bull. Jpn. Soc. Sci. Fish.*, 58, 547, 1992.
 66. Zhang, C.-H., Hirano, T., Suzuki, T., Shirai, T., and Morishita, A., Volatile compounds and their generation in smelts, *Bull. Jpn. Soc. Sci. Fish.*, 58, 773, 1992.
 67. Kasahara, K. and Nishibori, K., Volatile carbonyls and acids of smoked sardine, *Bull. Jpn. Soc. Sci. Fish.*, 48, 691, 1982.
 68. Komi, H., Kawai, T., and Ishida, Y., Flavor components of heated sardine (in Japanese), *Abstracts of Nippon Suisan Gakkai*, Spring Conference, Tokyo, 1992, 306.
 69. Kasahara, K. and Nishibori, K., Volatile components of roasted fishes, *Bull. Jpn. Soc. Sci. Fish.*, 51, 489, 1985.
 70. Fischetti, F. J., Natural and artificial flavors, in *CRC Handbook of Food Additives*, 2nd ed., Vol. 2, Furia, T. E., Ed., CRC Press, Boca Raton, FL, 1980, 295.
 71. German, J. B., Bruckner, G. G., and Kinsella, J. E., Lipoxygenase in trout gill tissue acting on arachidonic, eicosapentaenoic and docosahexaenoic acids, *Biochem. Biophys. Acta*, 875, 12, 1986.
 72. Gerwick, W. H., Bernart, M. W., Moghaddam, M. F., Jiang, Z. D., Solem, M. L., and Nagel, D. G., Eicosanoids from the Rhodophyta; new metabolism in the algae, *Hydrobiologia*, 204/205, 621, 1990.
 73. Nonaka, M., Hirata, F., Nakamura, M., and Sasamoto, Y., Volatile carbonyl compounds arising from highly nutritional fish meat for food stuff from sardine, *Bull. Jpn. Soc. Sci. Fish.*, 56, 67, 1990.
 74. Koizumi, C., Kieu-Thu, C.-T., and Nonaka, J., Undesirable odor of cooked sardine meat, *Bull. Jpn. Soc. Sci. Fish.*, 45, 1307, 1979.
 75. Nakamura, K., Iida, H., Tokunaga, T., and Miwa, K., Volatile flavor compounds of grilled red-fleshed fish: volatile carbonyl compounds and volatile fatty acids, *Bull. Jpn. Soc. Sci. Fish.*, 46, 221, 1980.
 76. McGill, A. S., Hardy, R., and Gunstone, F. D., Further analysis of the volatile components of frozen cold stored cod and the influence of these on flavor, *J. Sci. Food Agric.*, 28, 200, 1977.
 77. Matsuto, S., Nagayama, F., and Ono, T., Volatile monocarbonyls in frozen halibut, *Bull. Jpn. Soc. Sci. Fish.*, 33, 586, 1967.
 78. Angelini, P. and Merritt, C. J., Effect of irradiation on volatile constituents of stored haddock flesh, *J. Food Sci.*, 40, 197, 1975.
 79. Hebard, C. E., Flick, G. J., and Martin, R. E., in *Chemistry and Biochemistry of Marine Food Products*, Martin, R. E., Flick, G. J., Hebard, C. E., and Ward, D. R., Eds., AVI Publishing Company, Westport, CT, 1982, 149.
 80. Sakaguchi, M., Non-protein nitrogen compounds (of fish and shellfish), in *Micro Components of Fish and Shellfish* (in Japanese), Ikeda, S., Ed., Koseisha-Koseikaku, Tokyo, 1981, 2.
 81. Tokunaga, T., Studies on the quality evaluation of canned marine products: determination of the ratio of dimethylamine to trimethylamine in canned albacore, *Bull. Jpn. Soc. Sci. Fish.*, 41, 547, 1975.
 82. Shibamoto, T. and Russell, G. F., A study of the volatiles isolated from a D-glucose-hydrogen sulfide-ammonia model system, *J. Agric. Food Chem.*, 25, 109, 1977.
 83. Fujii, T., Hirayama, M., Okuzumi, M., Yasuda, M., Nishino, H., and Yokoyama, M., Shelf-life studies on fresh sardine packed with carbon dioxide-nitrogen gas mixture, *Bull. Jpn. Soc. Sci. Fish.*, 55, 1971, 1989.
 84. Kawai, T., Ishida, Y., and Higashida, T., Elimination of unpleasant odors of putrefied seafoods, *Bull. Jpn. Soc. Sci. Fish.*, 56, 1013, 1990.
 85. Burks, R. E., Baker, E. B., Clark, P., Esslinger, J., and Lacey, J. J. C., Detection of amines produced on irradiation of beef, *J. Agric. Food Chem.*, 7, 778, 1959.
 86. Tokunaga, T., Odor of fish and fish products, in *Odor of Marine Products*, Koizumi, C., Ed., Koseisha-Koseikaku, Tokyo, 1989, 21.
 87. Josephson, D. B., Lindsay, R. C., and Stuibler, D. A., Bisulfite suppression on fish aromas, *J. Food Sci.*, 48, 1064, 1983.
 88. Obatake, A., Tsumiyama, S., and Yamamoto, Y., Extractive nitrogenous constituents from the dark muscle of fish, *Bull. Jpn. Soc. Sci. Fish.*, 51, 1461, 1985.
 89. Hujita, M., Nitrogenous compounds of vertebrates, in *Extractive Components of Fish and Shellfish*, Sakaguchi, M., Ed., Koseisha-Koseikaku, Tokyo, 1988, 25.
 90. Daikoku, T. and Sakaguchi, M., Changes in level of trimethylamine and trimethylamine oxide during adaptation of young eel *Anguilla anguilla* to a seawater

- environment, *Bull. Jpn. Soc. Sci. Fish.*, 56, 1895, 1990.
91. Ooyama, S., Studies on the odor of canned fish: volatile components from the muscle of fresh fish, *Kaseigakuzasshi*, 24, 694, 1973.
 92. Whitaker, J., Development of flavor, odor, and pungency in onion and garlic, *Adv. Food Res.*, 22, 73, 1976.
 93. Ohtsuki, K., Kawabata, M., Taguchi, K., Kokura, H., and Kawamura, S., Determination of S-methyl-methionine, vitamin U, in various teas, *Agric. Biol. Chem.*, 48, 2471, 1984.
 94. Kawai, T., Irie, M., and Sakaguchi, M., Degradation of 2,4,6-trialkyltetrahydro-1,3,5-thiadiazines during storage, *J. Agric. Food Chem.*, 33, 393, 1985.
 95. Kubota, K., Kobayashi, A., and Yamanishi, T., Some sulfur containing compounds in cooked odor concentrate from boiled antarctic krills (*Euphasia superba* Dona), *Agric. Biol. Chem.*, 44, 2677, 1980.
 96. Werkhoff, P., Güntert, M., and Hopp, R., Dihydro-1,3,5-dithiazines: unusual flavor compounds with remarkable organoleptic properties, *Food Rev. Int.*, 8, 391, 1992.
 97. Kubota, K., Takeuchi, M., Kobayashi, A., Kitahara, T., and Ohashi, Y., Stereostructure of pyrrolidino[1,2-e]-4H-2,4-dimethyl-1,3,5-dithiazine, *J. Agric. Food Chem.*, 39, 1123, 1991.
 98. Hirai, Y., Fujii, M., Sakaguchi, S., Tano, J., Yanagihara, K., Katada, M., and Sakon, K., A study on live-human skin (in Japanese), *Abstr. Soc. Cosmet. Chem. Jpn.* (20th Meeting in Osaka), 23, 1985.
 99. Kawai, T. and Ishida, Y., Comparison of volatile components of dried squid to reaction products formed from the mixtures of hydrogen sulfide, ammonia, and aldehydes, *J. Agric. Food Chem.*, 37, 1026, 1989.
 100. Tanchotikul, U. and Hsieh, T. C.-Y., An improved method for quantification of 2-acetyl-1-pyrroline, a "popcorn"-like aroma, in aromatic rice by high-resolution gas chromatography/mass spectrometry/selected ion monitoring, *J. Agric. Food Chem.*, 39, 944, 1991.
 101. Tanaka, T., Aoki, Y., Tamase, K., Umoto, F., Ohbayashi, H., Imou, M., and Sasaki, M., Improved method for determination of total mercury and its application to vegetables and fruits in markets, *J. Food Hyg. Soc. Jpn.*, 33, 359, 1992.
 102. Komi, H., Kawai, T., and Ishida, Y., Formation mechanisms for heated food volatiles: features of animal and plant origin foods on the basis of change in pH of the distillates (in Japanese), *Abstr. Nippon Suisan Gakkai, Spring Conference* (Tokyo), 332, 1993.
 103. Hamada, J. S., Peptide glutaminase deamidation of proteins and protein hydrolysates for improved food use, *J. Am. Oil Chem. Soc.*, 68, 459, 1991.
 104. Izzo, H. V., Lincoln, M. D., and Ho, C.-T., Effect of temperature, feed moisture, and pH on protein deamidation in an extruded wheat flour, *J. Agric. Food Chem.*, 41, 199, 1993.
 105. Nishibori, K., Kanamitsu, S., and Okamoto, K., Flavor of the boiled short-necked clam (*Tapes (Amygdala) philippinarum*) and corbicula (*Corbicula japonica*), *Kaseigakuzasshi*, 23, 27, 1972.
 106. Whitfield, F. B., Volatiles from interactions of Maillard reactions and lipids, *Crit. Rev. Food Sci. Nutr.*, 31, 1, 1992.
 107. Hodge, J. E., Origin of flavor in foods: nonenzymatic browning reactions, in *Symposium on Foods: The Chemistry and Physiology of Flavors*, Schultz, H. W., Day, E. A., and Libbey, L. M., Eds., AVI Publishing Company, Westport, CT, 1967, 465.
 108. Hayashi, T., Ishii, H., Shinohara, A., Novel model experiment for cooking flavor research on crab leg meat, *Food Rev. Int.*, 6, 521, 1990.
 109. Fuke, S. and Konosu, S., Taste-active components in some foods: a review of Japanese research, *Physiol. Behav.*, 49, 863, 1991.
 110. Wöhler, F. and von Liebig, J., Thialdin and selenaldin: two synthetic organic bases, *Ann. Chem. Pharm.*, 61, 1, 1847.
 111. Aschan, O., Neues über aldehyd-ammoniak und chloral-ammoniak, *Chem. Ber.*, 48, 874, 1915.
 112. Brinkman, H. W., Copier, H., de Leuw, J. J. M., and Tjan, S. B., Components contributing to beef broth: analysis of the headspace volatiles of beef broth, *J. Agric. Food Chem.*, 20, 177, 1972.
 113. Choi, S. H., Kobayashi, A., and Yamanishi, T., Odor of cooked small shrimp, *Acetes japonicus* Kishinouye: difference between raw material and fermented product, *Agric. Biol. Chem.*, 47, 337, 1983.
 114. Kawai, T., Komi, H., and Ishida, Y., Formation mechanisms for dried short-necked clam volatiles upon heating (in Japanese), *Abstr. Nippon Suisan Gakkai, Autumn Conference* (Sanriku), 169, 1991.
 115. Buttery, R. G., Seifert, R. M., and Ling, L. C., Characterization of some volatile constituents of dry red beans, *J. Agric. Food Chem.*, 23, 516, 1975.
 116. Werkhoff, P., Brüning, J., Emberger, R., Güntert, M., Köpsel, M., and Kuhn, W., Studies on volatile sulfur-containing flavor components in yeast extract, in *Chemistry Analysis and Structure, Proceedings Vol. 4: 11th International Congress of Essential Oils, Fragrances and Flavors*, New Delhi, India, 1989, 215.
 117. Boelens, M., van der Linde, L. M., de Valois, P. J., van Dort, H. M., and Takken, H., Organic sulfur compounds from fatty aldehydes, hydrogen sulfide, thiols, and ammonia as flavor constituents, *J. Agric. Food Chem.*, 22, 1071, 1974.
 118. Bailey, M. E., The Maillard reaction and meat flavor, in *The Maillard Reaction in Food and Nutrition*, Waller, G. R. and Feather, M. S., Eds., ACS Symposium Series 215, American Chemical Society, Washington, D.C., 1983, 169.

119. **Shu, C. K., Mookherjee, B. D., Bandarovich, H. A., and Hagedorn, M. L.,** Characterization of bacon odor and other flavor components from the reaction of isovaleraldehyde and ammonium sulfide, *J. Agric. Food Chem.*, 33, 130, 1985.
120. **Shu, C. K., Hagedorn, M. L., Mookherjee, B. D., and Ho, C.-T.,** Volatile components of the thermal degradation of cystine in water, *J. Agric. Food Chem.*, 33, 438, 1985.
121. **Shu, C. K., Hagedorn, M. L., Mookherjee, B. D., and Ho, C.-T.,** pH effect on the volatile components in the thermal degradation of cysteine, *J. Agric. Food Chem.*, 33, 442, 1985.
122. **Zhang, Y., Chien, M., and Ho, C.-T.,** Comparison of the volatile compounds obtained from thermal degradation of cysteine and glutathione in water, *J. Agric. Food Chem.*, 36, 992, 1988.
123. **Schutte, L. and Koenders, E. B.,** Components contributing to beef flavor: natural precursors of 1-methylthio-ethanethiol, *J. Agric. Food Chem.*, 20, 181, 1972.
124. **Mecchi, E. P., Pippen, E. L., and Lineweaver, H.,** Origin of hydrogen sulfide in heated chicken muscle, *J. Food Sci.*, 29, 393, 1964.
125. **Fors, S.,** Sensory properties of volatile Maillard reaction products and related compounds: a literature review, in *The Maillard Reaction in Foods and Nutrition*, Waller, G. R. and Feather, M. S., Eds., ACS Symposium Series 215, American Chemical Society, Washington, D.C., 1983, 185.
126. **Collins, D. J. and Graymore, J.,** The basicity of dihydrothiazetidines, dihydrodithiazines, and tetrahydrothiadiazines, *J. Chem. Soc.*, 2893, 1958.
127. **Yamazaki, O. and Yoshinaka, T.,** Tables of standard components in foods (in Japanese), in *Official Data of Food Components Analyzed 4th. Edition*, Hitotsubashi-syuppan, Tokyo, 1992.